

A Variable OSR $\Sigma\Delta$ ADC using LSB-First SAR ADC as Quantizer

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by

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Abstract

The demand for high-resolution, energy-efficient, moderate-speed data converters has been continuously increasing in modern electronics devices. Successive Approximation Register (SAR) Analog-to-Digital Converter is adopted because it offers low power consumption and scalability to advanced CMOS technologies. However, the maximum achievable resolution is limited to 12-14 bits by the non-linearity of the internal Digital-to-Analog Converter (DAC) and comparator offset [1]. Noise-Shaping SAR ADC has emerged as a promising solution to overcome this limitation, which integrates Delta-Sigma Modulator (DSM) and SAR ADC architecture together to form a multibit noise-shaping hybrid architecture. This combines the advantages of both techniques, resulting in high resolution while operating at moderate bandwidths.

Signal-to-Quantization Noise Ratio (SQNR) of an oversampled Analog-to-Digital Converter (ADC) like Noise-Shaping SAR ADC, is determined by sampling frequency, input signal bandwidth, and the resolution of the quantiser in the loop. The sampling frequency is limited by the latency of the quantiser in the loop, and non-linearities limit resolution. Beyond a certain limit, it is necessary to reduce the input signal bandwidth to improve the SQNR. This work presents a Non-Uniform sampling technique to improve the latency of the ADC, allowing for better Signal-to-Noise Ratio (SNR) to be achieved at the same input signal bandwidth and resolution. This work uses a different search algorithm for Successive Approximation (SA) called LSB-First SA for converging to the required digital output. This algorithm takes 2 to $2N+1$ cycles for conversion and will stay idle for the whole $2N+1$ cycles even if conversion completes earlier. This work presents a modified version of the LSB-First SAR ADC to eliminate the idle time and improve the sample density. This Variable-rate architecture is used as a quantiser in a Delta-Sigma ADC to further improve the SQNR of the ADC. Simulation results based on an 8-bit implementation indicate that the proposed method achieves an approximately 10 dB enhancement in SQNR for first-order variable Oversampling Ratio (OSR) noise-shaping ADCs compared with fixed OSR conventional noise-shaping architecture.

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Abbreviations

SAR Successive Approximation Register.

DAC Digital-to-Analog Converter.

DSM Delta-Sigma Modulator.

SQNR Signal-to-Quantization Noise Ratio.

ADC Analog-to-Digital Converter.

SNR Signal-to-Noise Ratio.

SA Successive Approximation.

OSR Oversampling Ratio.

FFT Fast Fourier Transform.

NU-DFT Non-Uniform Discrete Fourier Transform.

TPMS Tire Pressure Monitoring Systems.

FIR Finite Impulse Response.

LTI Linear-Time Invariant.

MSB Most Significant bit.

CDAC Charge sharing DAC.

LSB Least Significant bit.

AAF Anti Aliasing Filter.

PE Priority Encoder.

LTV Linear-Time Variant.

ENOB Effective Number of Bits.

CIFB Cascade Integrator Feedback.

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Chapter 1

Introduction and Literature review

Analog-to-Digital Converter (ADC) is a circuit that is used to convert a continuous-time signal to digital. Modern-day computations primarily rely on digital systems due to their superior robustness to noise and process variations, along with their ability to enable scalable and error-tolerant information processing, supporting operations that are near impossible for analog systems. Since all real-world signals naturally exist in analog form, it is necessary to have a system that can interface the analog world with the computation domain, and that is where ADC comes into the picture.

ADC is designed for various applications that require high-speed operation, power-efficient conversion, or high-resolution signal conversion, depending on the characteristics of the input signal and the overall constraints of the system. In systems demanding high throughput, such as broadband wireless receivers, software-defined radios, and high-speed serial links, the ADC must operate at very high sampling frequencies while preserving signal integrity and minimizing distortion. These applications prioritize speed, bandwidth, and latency over resolution.

However, applications such as sensor readout circuits, medical imaging, and precision measurement systems demand high accuracy and low noise performance. In such cases, the ADC is optimized for high resolution, excellent signal-to-noise ratio (SNR), and low power consumption, often at lower sampling rates. The choice of ADC architecture - such as Flash, SAR, pipelined, Dual-Slope ADC, or noise-shaping ADCs - is therefore strongly influenced by the operational speed, achievable resolution, power efficiency, and implementation complexity, which is reviewed in detail in the upcoming section.

In this thesis, we review different architectures of ADCs and discuss their suitability for different speed resolution combinations. We then proposed a new architecture for speed resolu-

tion combinations that is challenging to achieve using conventional architectures.

1.1 ADC architectures

Over the decades, ADCs have evolved significantly, driven by diverse requirements of modern electronic systems. Designers focused on the target requirement of the ADC specifications from different application point of view and came up with various architectures. The design ideas focused on improving the speed, power efficiency, or accuracy of the architecture. These widely used ADC architectures are categorized into Nyquist-rate and oversampling converters. In this Section, we will look at a few architectures that will give the background for the work in the thesis.

1.1.1 Nyquist-rate converter I : Flash ADC

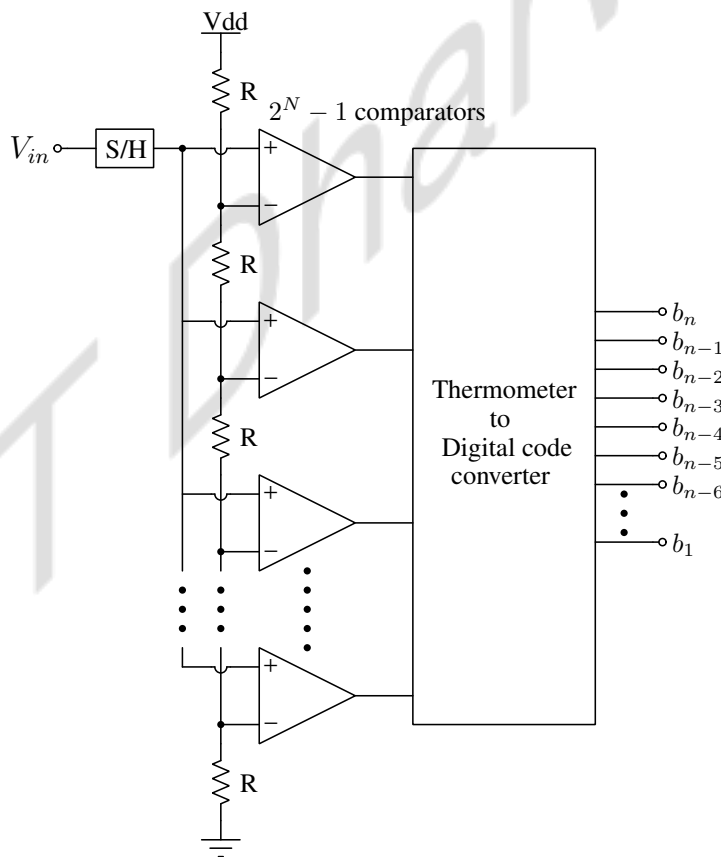


Figure 1.1: N-bit Flash ADC [2]

The Flash ADC, in particular, is built for applications that require high-speed operations. The architecture consists of an array of comparators that operate in parallel, leading to low

latency as shown in Figure 1.1. In this architecture, the number of comparators doubles with the addition of every bit, which will increase the area exponentially. Static and dynamic power dissipation in the architecture are due to the presence of a resistor and a comparator array, respectively, which increase with resolution. The input capacitance also increases exponentially with the resolution.

Comparators' offsets is a major concern in high-resolution Flash ADC as the LSB step size decreases with increasing resolution. Calibrating the comparators adds further overhead in terms of area, power, and complexity. Hence, Flash ADC is preferred for use cases emphasizing speed rather than resolution (typically ≤ 8 bits [1]).

1.1.2 Nyquist-rate converter II: Successive Approximation Register ADC

Not all applications demand high-speed operation. To deal with the applications where moderate speed would be sufficient, and power efficiency is crucial, Successive Approximation Register (SAR) ADC is generally preferred. SAR ADC does serial operations by taking a few iterations to converge to the required digital output. It is a single comparator architecture consisting of a feedback Digital-to-Analog Converter (DAC) and a digital logic (SAR) to perform the iterations (successive approximations) as shown in Figure 1.2. The output of ADC is computed by iteratively narrowing the search space using binary search, the steps of which constitute the final approximation.

Since SAR ADC heavily relies on the accuracy of DAC and the comparator, non-linearity in DAC and comparator offset limit the resolution of SAR ADC to 12-14 bits [1]. SAR ADC remains a widely preferred architecture due to its power efficiency and scalability.

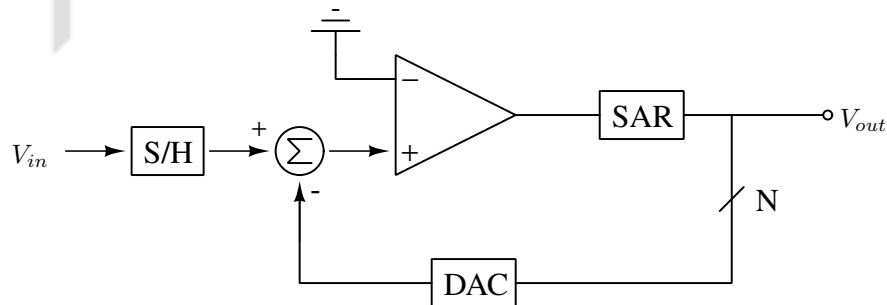


Figure 1.2: N-bit SAR ADC [2]

We have surveyed to review and compare the performance of Flash and SAR ADCs, as illustrated in Figure 1.3. The data for this plot has been taken from the ADC Performance Survey

by Boris Murmann [1]. The comparative analysis enables conclusions to be drawn regarding the suitability of each ADC architecture for different application domains, based on key performance metrics such as achievable resolution, power consumption, and operating bandwidth. From Figure 1.3(a), it is evident that SAR ADC offers significantly better power efficiency compared to Flash ADC while achieving better resolution. While trying to achieve high resolution, the conversion speed of SAR ADC drops down significantly as shown in Figure 1.3(b).

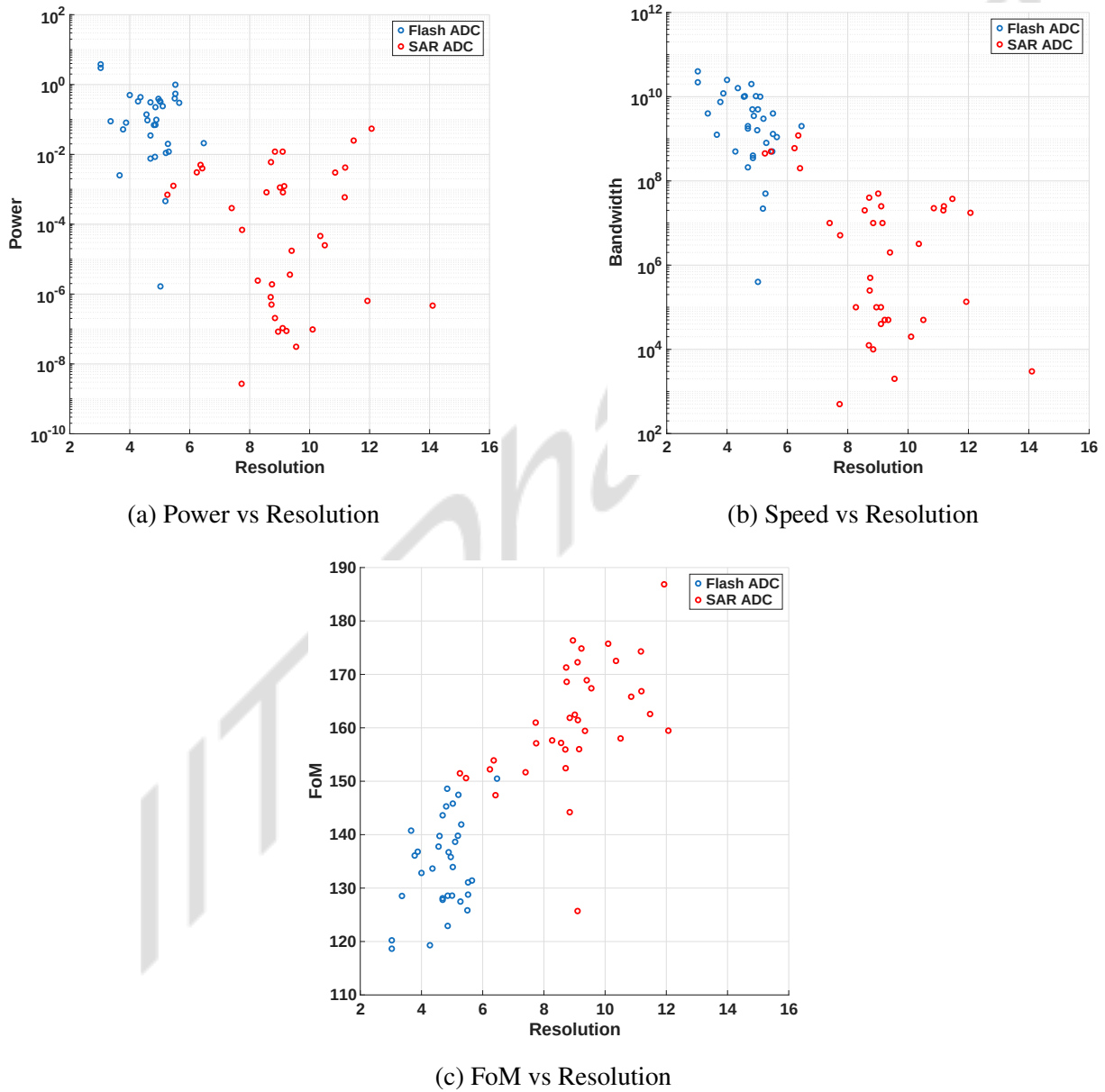


Figure 1.3: Comparison of Flash ADC and SAR ADC w.r.t their achievable resolution, power consumption, bandwidth, and Figure-of-Merit ($\text{FoM} = \text{SNR} + 10 \log(\text{Bandwidth}/\text{Power})$)

1.1.3 Nyquist-rate converter III : Pipelined ADC

To overcome the speed limitations of SAR ADC while still retaining its low power nature and scalability, a pipelined architecture was developed. It consists of multiple stages, each consisting of sub-ADC, DAC, and a residue amplifier with N bits divided among the stages as shown in Figure 1.4. The sub-ADCs can be chosen to be SAR ADC or Flash as per the design requirements.

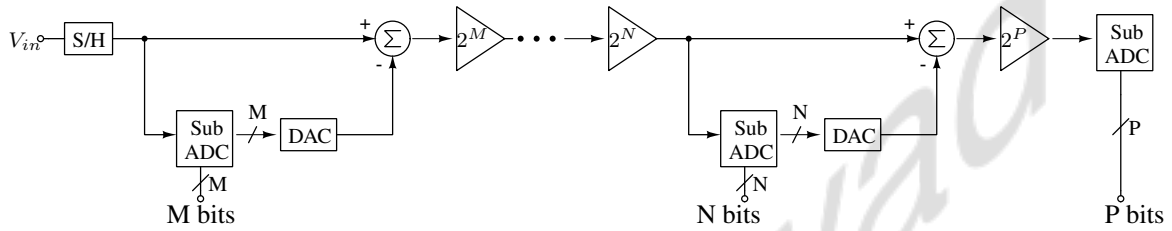
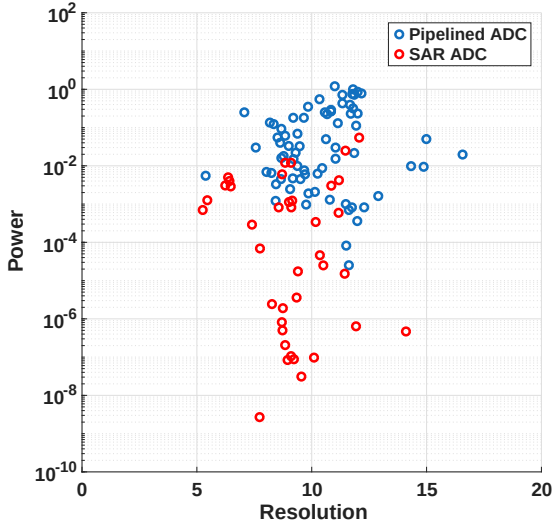


Figure 1.4: Pipeline ADC [2]

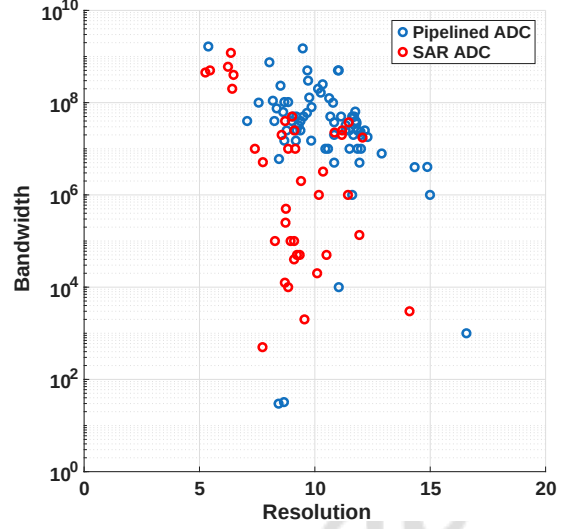
A conventional N-bit Flash ADC employs 2^N comparators, while a pipelined Flash ADC achieves a substantial reduction in comparator count by partitioning the conversion into multiple stages. However, this architectural simplification introduces additional latency, as each stage operates sequentially. These will add up to the total delay of the Pipelined ADC. Here, each stage consists of a ADC, DAC, and a residue amplifier contributing to the delay of each stage.

Pipelined ADC is essentially a feedforward style implementation of SAR algorithm. This trades latency to gain speed at the expense of additional hardware. Each stage demands accuracy, as errors in the early stages will accumulate in the later stages. So, calibration of blocks and addition of a redundant bit for every stage is done to maintain accuracy, which will increase the complexity and area. Pipelined ADC is a good choice for applications requiring moderate-high speed performance while achieving a resolution of 10-14 bits [3].

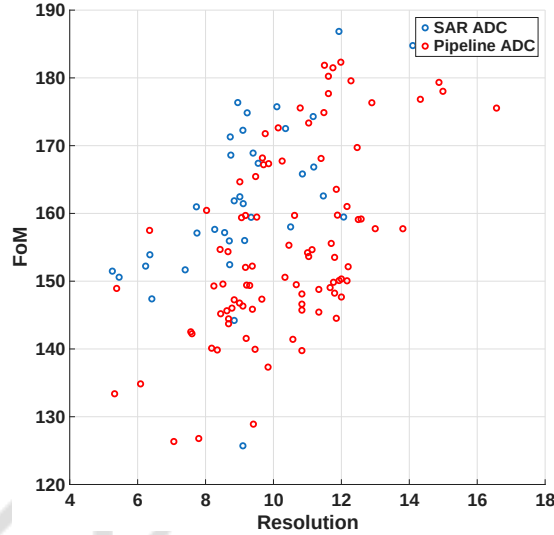
We have surveyed to review and compare the performance of Pipelined ADC and SAR ADCs, as illustrated in Figure 1.5. The data for this plot has been taken from the ADC Performance Survey by Boris Murmann [1]. The comparative analysis enables conclusions to be drawn regarding the suitability of each ADC architecture for different application domains, based on key performance metrics such as achievable resolution, power consumption, and operating bandwidth. From Figure 1.5(a), it is evident that SAR ADC offers significantly better power efficiency compared to Pipelined ADC. From Figure 1.5(b), it is evident that the pipelined ADC has better bandwidth compared to the SAR ADC.



(a) Power vs Resolution



(b) Speed vs Resolution



(c) FoM vs Resolution

Figure 1.5: Comparison of Pipelined ADC and SAR ADC w.r.t. their achievable resolution, power consumption, bandwidth, and Figure-of-Merit ($\text{FoM} = \text{SNR} + 10 \log(\text{Bandwidth}/\text{Power})$)

1.1.4 Oversampling converter: Delta-Sigma ($\Delta\Sigma$) ADC

The Signal-to-Noise Ratio (SNR) of Nyquist ADCs is limited by resolution and latency, making it nearly impossible for Nyquist rate converters to achieve the resolution equivalent to 16 bits or higher [4]. To overcome these limitations, Delta-Sigma Modulator (DSM) was introduced. Unlike Nyquist converters, it uses low resolution ADC (even as low as 1-bit) as a quantiser shown in Figure 1.6 to achieve resolutions greater than 16 bits.

DSM achieves this through the combined use of oversampling and noise shaping. It oversamples the signal to reduce the quantization noise inside the signal band due to the increase in the sampling clock bandwidth (across which the noise floor of the ADC is spread) and attenuates the noise in lower frequency bands to reduce it further. Hence, it is preferred for applications requiring excellent noise immunity and dynamic range.

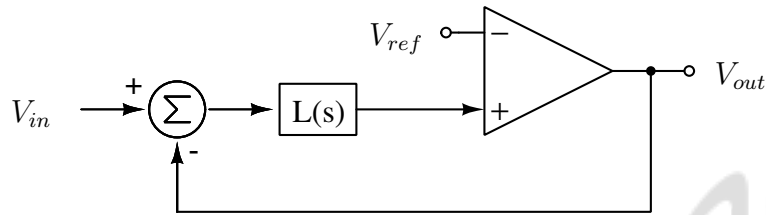


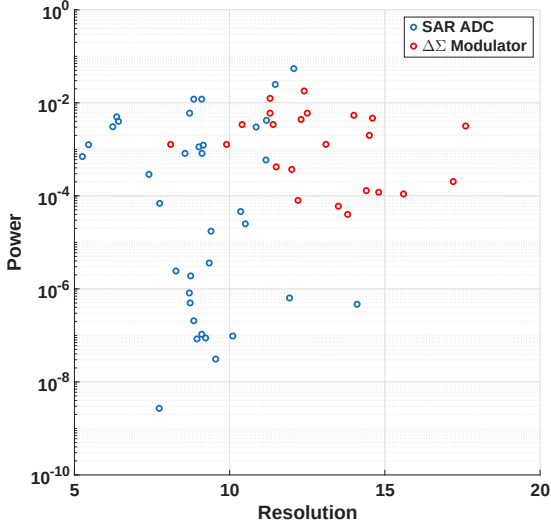
Figure 1.6: Delta-Sigma Modulator [5]

We have surveyed to review and compare the performance of DSM and SAR ADCs, as illustrated in Figure 1.7. The data for this plot has been taken from the ADC Performance Survey by Boris Murmann [1]. The comparative analysis enables conclusions to be drawn regarding the suitability of each ADC architecture for different application domains, based on key performance metrics such as achievable resolution, power consumption, and operating bandwidth. From Figure 1.7(b), it can be concluded that a 1-bit $\Delta\Sigma$ Modulator manages to achieve a resolution better than a multibit SAR ADC. This improvement in resolution is primarily enabled by oversampling and noise shaping, which push quantization noise out of the input signal bandwidth. However, this benefit is obtained at the cost of input signal bandwidth, which is a tradeoff in oversampling converters like DSM.

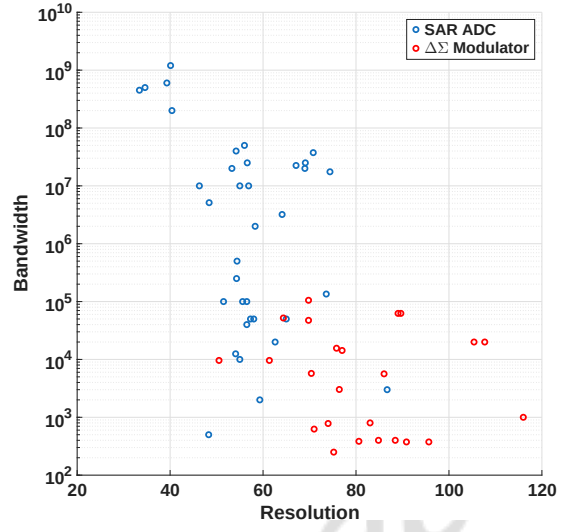
To address the bandwidth limitations associated with the 1-bit DSM, a multibit SAR ADC can be used as a quantiser for DSM. This leads to the realization of a hybrid noise-shaping SAR ADC architecture, as illustrated in Figure 1.8. This will combine the advantages of both architectures. The key advantages of this hybrid architecture include-

1. Excellent noise immunity and dynamic range of DSM.
2. Broader input bandwidth and power-efficient nature of SAR ADC.

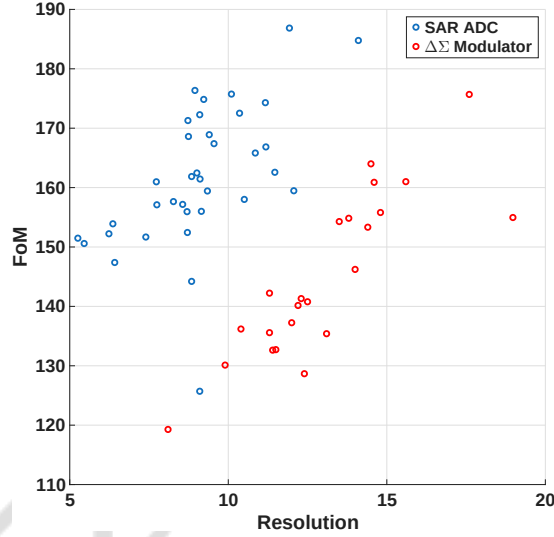
As a result, the proposed architecture achieves high resolution and strong noise shaping without significantly curtailing input bandwidth, thereby overcoming the fundamental limitations of conventional single-bit $\Delta\Sigma$ modulators.



(a) Power vs Resolution



(b) Speed vs Resolution



(c) FoM vs Resolution

Figure 1.7: Comparison of SAR ADC and 1-bit Delta-Sigma Modulator w.r.t. their achievable resolution, power consumption, bandwidth, and Figure-of-Merit ($\text{FoM} = \text{SNR} + 10 \log(\text{Bandwidth}/\text{Power})$)

We have surveyed to review and compare the performance of different ADC architectures, as illustrated in Figure 1.9. The data for this plot has been taken from the ADC Performance Survey by Boris Murmann [1] and the overview and survey papers [6]. The comparative analysis enables conclusions to be drawn regarding the suitability of each ADC architecture for different application domains, based on key performance metrics such as achievable resolution, power consumption, and operating bandwidth. The combined benefit of DSM and SAR ADC in the noise-shaping SAR ADC is clearly demonstrated in Figure 1.9, which highlights the

improved trade-off between resolution and bandwidth offered by the noise-shaping SAR ADC.

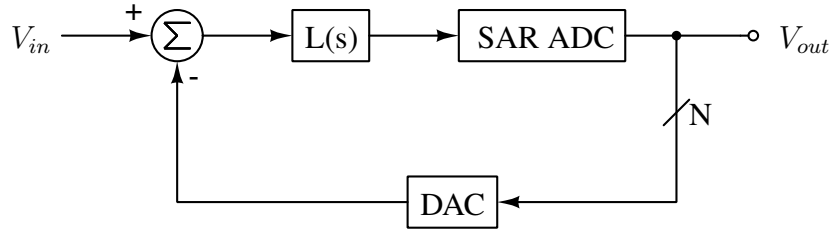
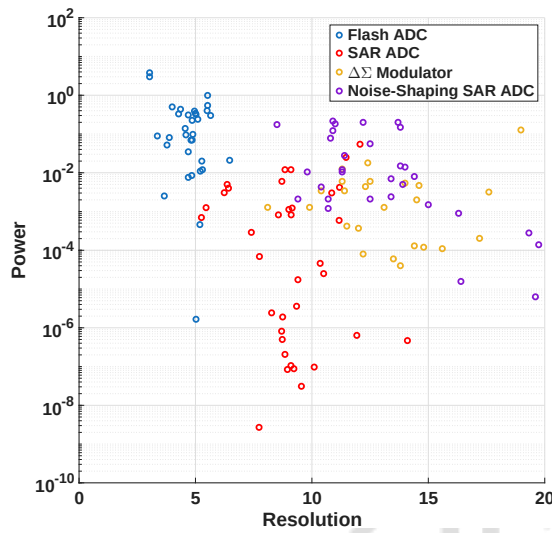
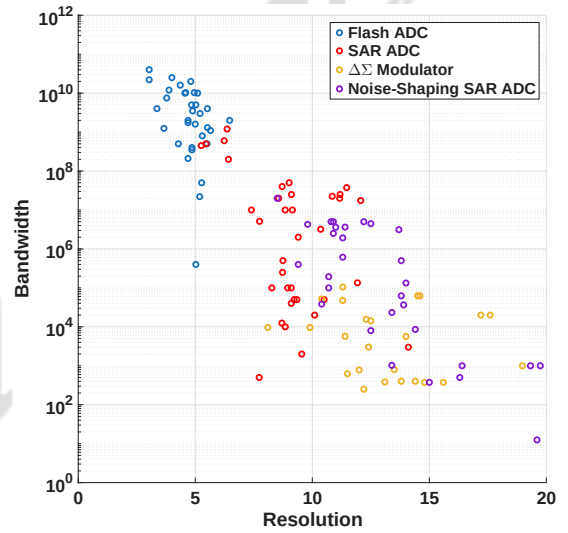


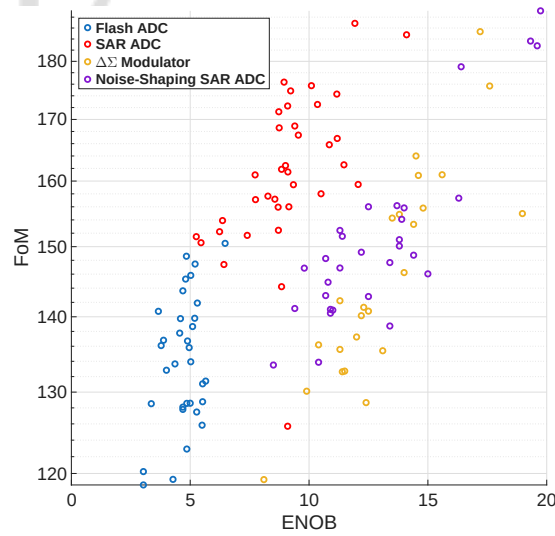
Figure 1.8: Noise-Shaping SAR ADC [5]



(a) Power vs Resolution



(b) Speed vs Resolution



(c) FoM vs Resolution

Figure 1.9: State-of-Art of different ADC architectures

1.2 Major contributions

This work investigates core ADC architectures, their limitations, and motivates the development of Noise-Shaping SAR ADC. Additionally, it details SAR ADC and its variant called LSB-First SAR ADC, and the motivation behind the introduction of the same. The contributions of this thesis are as follows:

1. This work also introduces a concept of non-uniform sampling ADC and details how it is useful in improving the performance of ADC. The work details the use of variable Over-sampling Ratio (OSR) LSB-First SAR ADC to demonstrate effective OSR improvement.
2. This work also details the use of variable latency ADC in an oversampling Noise-shaping converter to achieve higher effective SNR by 1st and 2nd noise shaping techniques.

1.3 Thesis outline

The thesis is organized as follows.

1. Chapter 2 discusses the basic concepts of ADC, prior art to this work, and the problem statement.
2. Chapter 3 discusses the methodology adopted to solve the problem statement discussed in Chapter 2. It also describes modeling strategies and system-level design of ADC architecture.
3. Chapter 4 discusses the Variable OSR $\Delta\Sigma$ Modulator and system design of multibit noise-shaping using ADC architecture discussed in Chapter 3 as quantiser.
4. Finally, Chapter 5 presents the conclusions and outlines directions for future work.

Chapter 2

ADC basics, Prior art and Development

2.1 Basic concepts of Analog to Digital Converter (ADC)

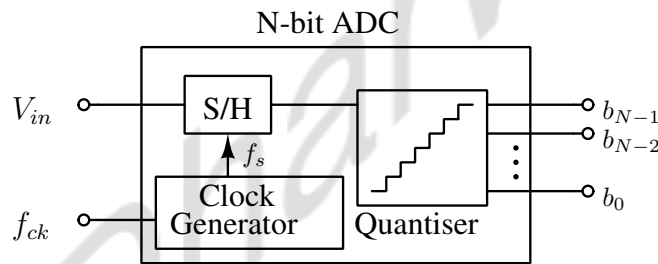


Figure 2.1: Block Diagram of ADC architecture

N-bit ADC consists of a sample and hold circuit followed by a quantiser as shown in Figure 2.1. Quantiser consists of 2^N quantization levels. ADC samples a continuous-time signal and maps it to a quantization level (digital data). The sampling speed is limited by the “**latency**” of ADC in a single-stage architecture, i.e, conversion delay of ADC to convert analog into a digital signal. While the maximum sampling frequency is limited by bandwidth, the minimum sampling frequency is dependent on the input signal bandwidth, as the Nyquist criterion should be followed to avoid aliasing.

2.1.1 Sampling and Aliasing in bandlimited signals

A signal of bandwidth f_{in} creates replicas of it when sampled, centered at f_s , $2f_s$, ... in the frequency domain as shown in Figure 2.2.

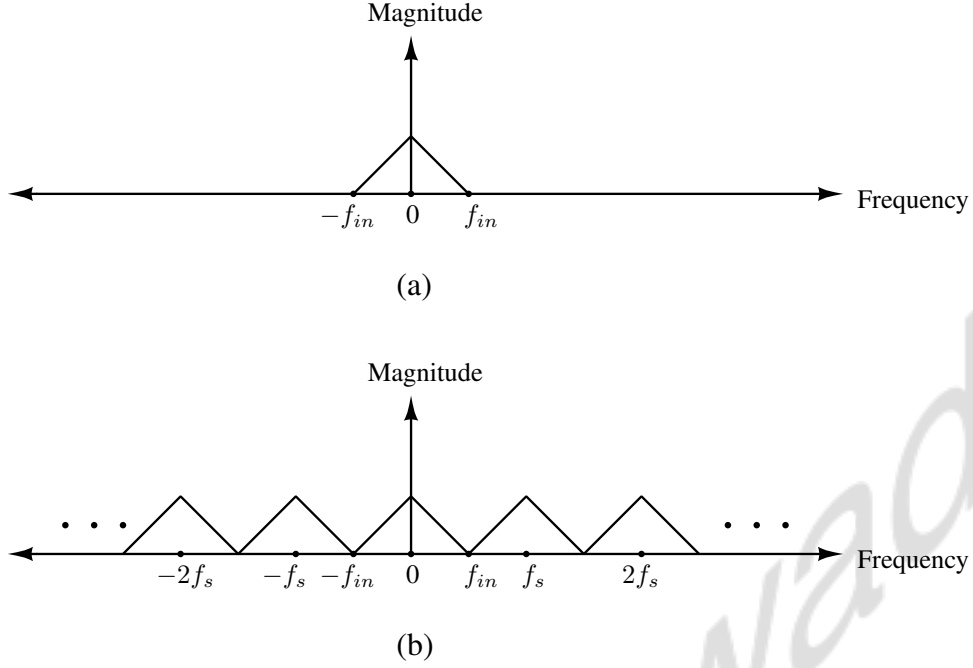


Figure 2.2: Spectrum of (a) Bandlimited signal (b) Sampled signal

Spectral Replication Due to Sampling

Uniform sampling of $x(t)$ with sampling period $T_s = 1/f_s$ can be modeled as multiplication by a periodic impulse train:

$$x_s(t) = x(t) \cdot p(t), \quad (2.1)$$

where

$$p(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_s). \quad (2.2)$$

The Fourier transform of the impulse train $p(t)$ is given by

$$P(f) = \frac{1}{T_s} \sum_{k=-\infty}^{\infty} \delta(f - kf_s). \quad (2.3)$$

Since multiplication in the time domain corresponds to convolution in the frequency domain, the spectrum of the sampled signal is

$$X_s(f) = X(f) * P(f). \quad (2.4)$$

Substituting for $P(f)$ yields

$$X_s(f) = X(f) * \left(\frac{1}{T_s} \sum_{k=-\infty}^{\infty} \delta(f - kf_s) \right). \quad (2.5)$$

Using the convolution property with delta functions, this simplifies to

$$X_s(f) = \frac{1}{T_s} \sum_{k=-\infty}^{\infty} X(f - kf_s) \quad (2.6)$$

Equation 2.6 shows that uniform sampling results in an infinite number of replicas of the original spectrum $X(f)$, each shifted by integer multiples of the sampling frequency f_s and scaled by $1/T_s$. These spectral replicas are centered at

$$f = kf_s, \quad k \in \mathbb{Z}. \quad (2.7)$$

If the signal bandwidth satisfies $B < f_s/2$, the replicated spectra do not overlap and perfect reconstruction of the original signal is possible. Otherwise, spectral overlap occurs, resulting in aliasing.

When a signal is sampled at frequency $f_s = 2f_{in}$, it is called Nyquist rate sampling. It is the minimum required sampling frequency for proper reconstruction of the signal after sampling. If the signal is sampled at a rate less than the Nyquist rate, it is called undersampling. This will lead to aliasing as shown in Figure 2.3. Aliasing leads to noise folding of high-frequency components into low-frequency bands, which distorts the signal and changes the period of the signal in the case of periodic signals. The original signal cannot be reconstructed from the sampled signal.

The Nyquist rate-sampled signal requires a brick wall or an ideal filter for proper reconstruction, as shown in Figure 2.4. Since a practical filter cannot be a brick wall, it distorts the signal and does not attenuate out of band components completely. To circumvent these, Nyquist-rate ADCs generally sample at $f_s > 2f_{in}$ to accommodate the filter roll-off as shown in Figure 2.5.

2.1.2 Quantization noise in ADC

The analog input consists infinite number of possible values while the output consists of 2^N quantization levels in an N-bit ADC. Due to this, quantization noise will be seen in the ADC, which is given by

$$Q_e = V_{in,s} - V_{out}$$

where,

1. $V_{in,s}$ is the sampled input.

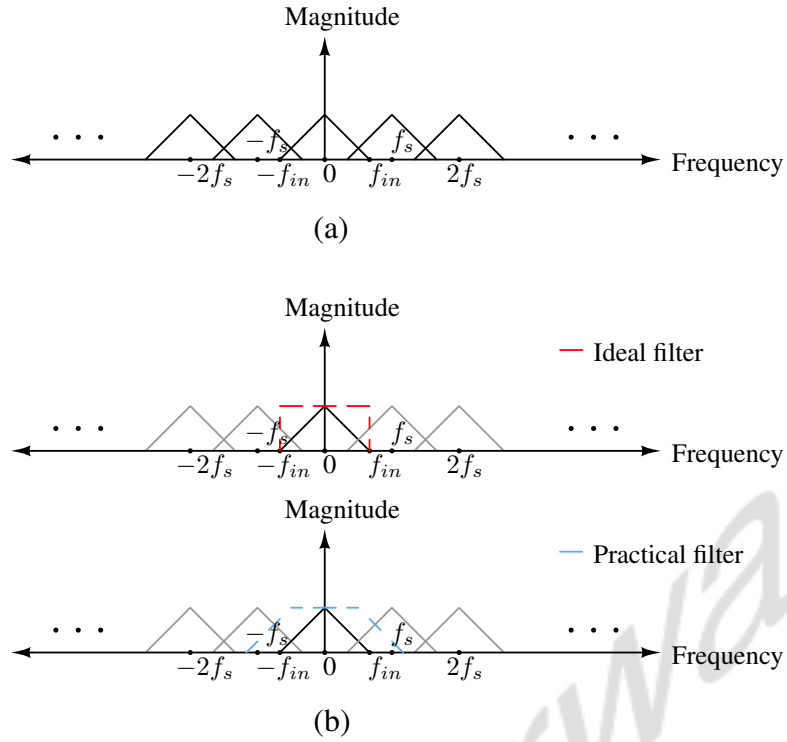


Figure 2.3: Spectrum of (a) Under-sampled signal (b) Reconstructed undersampled signal

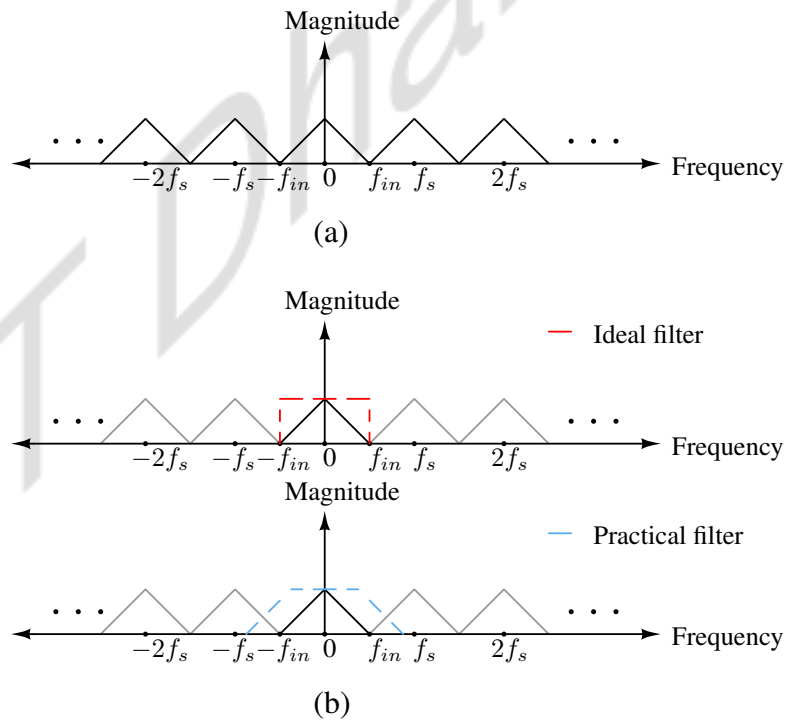


Figure 2.4: Spectrum of (a) Nyquist rate sampled signal (b) Reconstructed Nyquist rate sampled signal

2. V_{out} is analog equivalent of digital output.

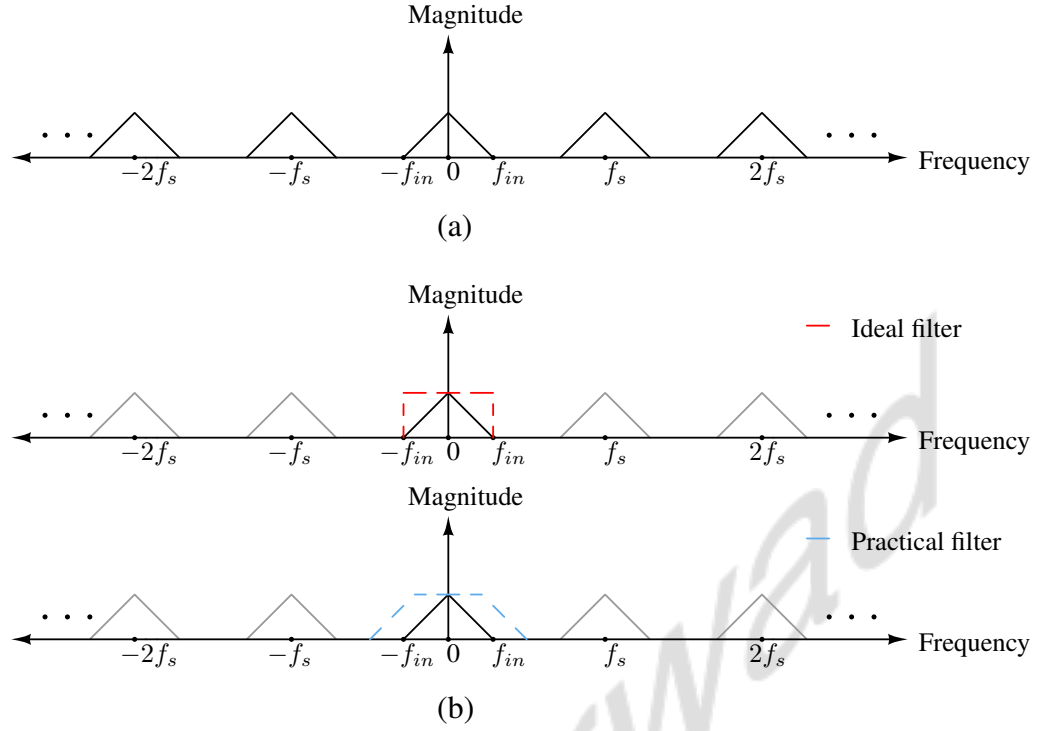


Figure 2.5: Spectrum of (a) Oversampled signal (b) Reconstructed oversampled signal

Since the output of ADC consists of quantization noise, Figure 2.1 can be modeled as shown in Figure 2.6. The smallest change in input voltage that produces a change of one LSB

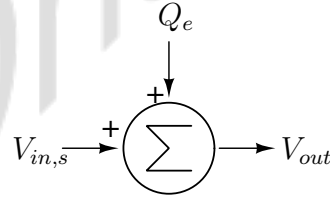


Figure 2.6: Quantization noise model of ADC

at the digital output is called the “**step size**” of ADC. Step size of ADC is given by

$$\Delta = \frac{V_{ref}}{2^N}$$

V_{ref} is the reference voltage of ADC and Δ can also be defined as the required change in sampled input to observe a change in ADC output.

2.1.3 Quantization noise power

Quantization noise is not predictable unless the input is known. It is dependent on the nature of the input signal to the ADC. To derive the quantization noise power, it is necessary to make a

few assumptions. We consider the following assumptions.

1. We assume an ideal mid-tread¹ quantiser with step size Δ .
2. Quantization noise Q_e uniformly distributed in the interval $[-\Delta/2, \Delta/2]$ as shown in Figure 2.7.

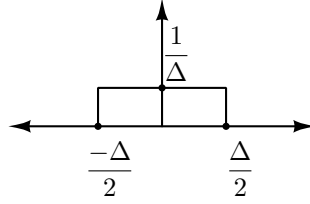


Figure 2.7: Probability density function for ADC

Quantization noise power is given by

$$\sigma^2 = E\{e^2\} = \int_{-\Delta/2}^{\Delta/2} e^2 p(e) de = \frac{1}{\Delta} \int_{-\Delta/2}^{\Delta/2} e^2 de$$

where $p(e)$ is the probability density function.

$$\sigma^2 = \frac{\Delta^2}{12} \quad (2.8)$$

This is the total quantization noise power across the band 0 to $\frac{f_s}{2}$, as shown in Figure 2.8, considering the uniform error model and f_s as the Nyquist rate.

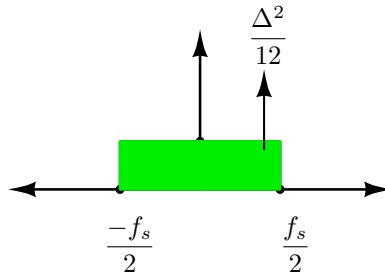


Figure 2.8: Noise-Power spectral density for a Nyquist-rate ADC with area under the curve equal to $\frac{\Delta^2}{12}$

¹A mid-tread quantizer is a type of uniform quantizer in which the zero input level lies at the center of a quantization interval, rather than at a decision threshold [4].

SNR of ADC

SNR of ADC is the ratio of signal power to the noise power. Considering we use an ideal filter to reconstruct the signal, SNR is given by

$$\text{SNR} = \frac{\text{Signal Power}}{\text{Noise power inside signal band}}$$

SNR can be improved by decreasing the noise power. This can be achieved through the following methods.

1. Total quantization power is a function of the step size of ADC, which is evident from Equation 2.10. Increasing resolution will decrease noise power.
2. For a fixed resolution, total quantization noise power remains constant. Increasing sampling frequency will spread the power across the clock bandwidth, thereby decreasing the magnitude of noise power in the signal bandwidth.

2.1.4 Dynamic range (DR) of ADC

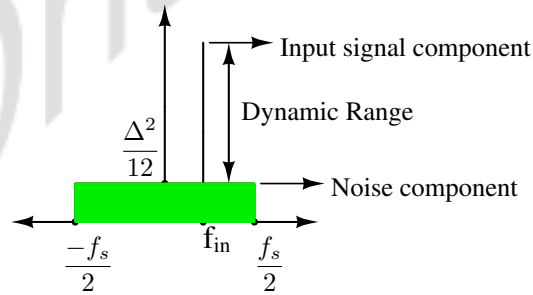


Figure 2.9: Power spectral density for a Nyquist-rate ADC

The ratio of the largest and the smallest signal amplitude that ADC can measure is the “dynamic range” of the ADC, given by the following equation:

$$DR = 10 \log_{10} (P_{\text{signal,max}} / P_{\text{noise}}) \text{ dB}$$

The dynamic range (DR) of an ADC is the ratio between the largest non-saturating input signal and the smallest detectable input signal. It quantifies how well the ADC can distinguish small signals in the presence of noise while still accommodating large signals without distortion or saturation.

2.2 Prior art

Chapter 1 provided an overview and development of widely used ADC architectures over the decades. Building on that foundation, this section focuses on operating principles of ADC which are widely preferred in systems where SNR has the highest priority.

2.2.1 Dual-Slope ADC

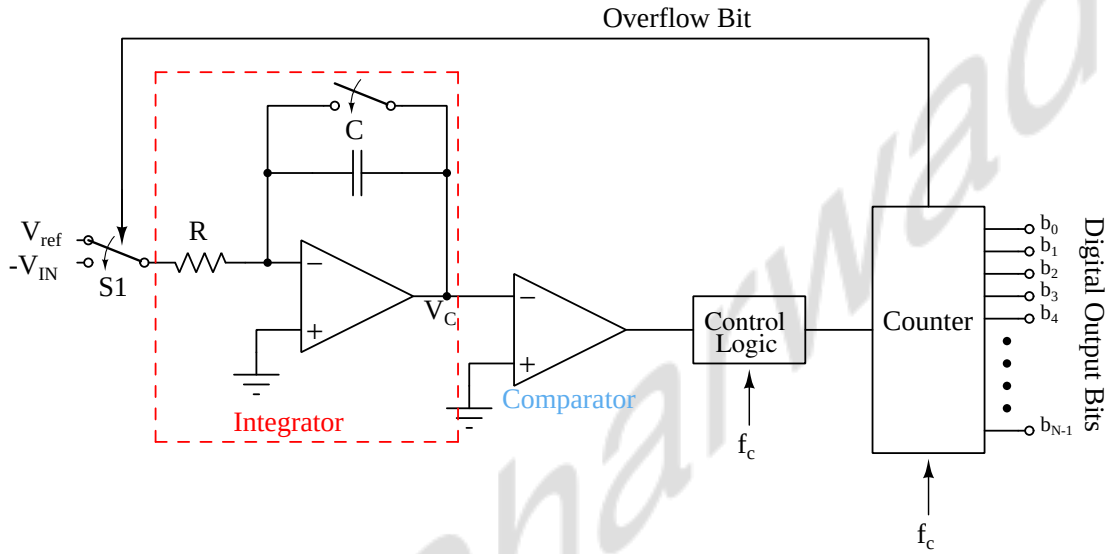


Figure 2.10: Dual Slope ADC [2]

Dual slope ADC, also known as integrating ADC, is shown in Figure 2.10 [2]. An integrating type ADC converts an analog input into a digital value by measuring time, rather than amplitude alone. The core idea is to accumulate the input signal over a defined interval and then relate this accumulated value to a digital count. This approach offers excellent noise rejection and accuracy.

It consists of an integrator, a comparator, an upcounter, and a digital logic to control the counter [2]. It works in two phases. In the first phase, the integrator is connected to the input signal, accumulating the input value across the capacitor C , causing the voltage across it to ramp up. Next phase, it is connected to the DC signal, V_{ref} , resulting in the integrator output going to zero.

V_A and V_B are 2 different input samples shown in Figure 2.11. Phase 1 and Phase 2 are charging and discharging phases, respectively. The following is the working of the Dual-Slope ADC, divided into 2 phases.

Phase 1 (Charging phase) :

This phase starts by charging the capacitor, C, to V_{IN} . The integrator output reaches

$$V(T_{up}) = \frac{V_{in}}{RC} T_{up}$$

This results in the first slope shown in Figure 2.11. Due to the voltage across the capacitor, C, there will be a transition in the comparator output, triggering the counter to do the up-counting till it overflows. Once the counter overflows, the overflow bit is triggered. The switch S1 controlled by the overflow bit switches the input of the integrator from V_{IN} to V_{ref} . This marks the end of phase 1. To trigger the overflow bit, the counter takes 2^N cycles. So, the charging phase is of fixed length irrespective of the input V_{IN} .

$$T_{up} = T(2^N \text{ cycles})$$

Phase 2 (Discharging phase) :

This phase starts when the overflow bit of the counter is triggered. The switch S1 is connected to the fixed reference voltage V_{ref} due to which the capacitor C starts to discharge, resulting in the second slope shown in Figure 2.11. The counter does the up-counting till the capacitor completely discharges. When the capacitor completely discharges, there will be a transition in the comparator output again, due to which upcounting is stopped, and the necessary digital output is held. As the integrator is connected to a fixed source, the discharge rate across the capacitor is constant. Depending on the voltage across the capacitor, C, the time taken to discharge varies. Time taken to discharge and to converge the digital output to the required value is T_{down}

$$T_{down} = \frac{V_{in}}{V_{ref}} T_{up}$$

The following conclusions are drawn from the above ADC architecture

1. Conversion time = $T_{up} + T_{down} = 2^N(\text{fixed}) + 2^N$ (variable as per input data)
2. Conversion time is variable and is very slow, ranging from 2^N+1 to 2^{N+1} .
3. Dual-Slope ADC is known for its high performance when it comes to SNR [2]. But looking at the conversion time, it is evident that the latency of the architecture is very high while they try to achieve excellent noise rejection and accuracy.

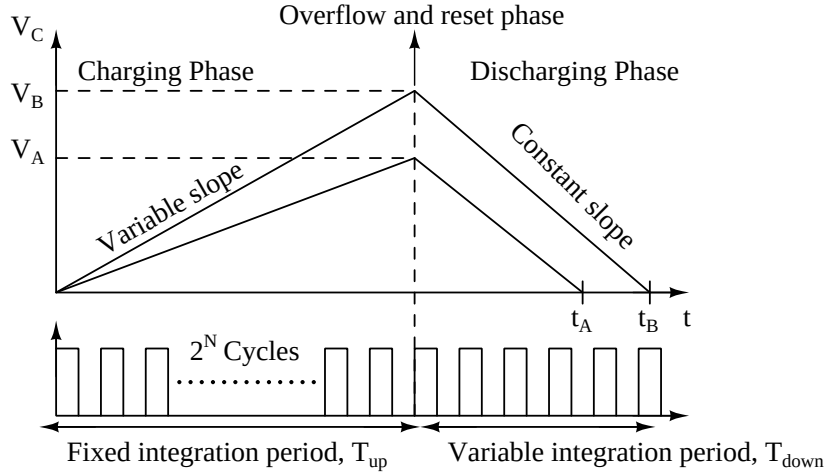


Figure 2.11: Working of Dual Slope ADC in different phases [2]

2.2.2 Successive Approximation Register ADC (SAR ADC)

SAR ADC converts an analog input into a digital code by performing a binary search on the input voltage. It is an iterative method in which the input signal is compared against a sequence of internally generated reference levels, refining the estimate one bit at a time, starting from the most significant bit (MSB) and ending at the least significant bit (LSB).

The SAR ADC converges into the desired output via a binary search operation over the input range $[0, V_{ref}]$. Each comparison halves the remaining uncertainty. During each comparison cycle, the evaluation is performed to determine whether the input voltage exceeds or falls below a DAC-generated reference corresponding to a previous digital code. These decisions effectively divide the remaining voltage range by a factor of two on every iteration, thereby narrowing the quantization error associated with the input estimate.

With every successive comparison/iteration, one additional bit of the output code is resolved, and the maximum possible quantization error is reduced accordingly. After N comparison steps, the quantization error in the input voltage is limited to:

$$\text{Resolution after } N \text{ steps} = \frac{V_{ref}}{2^N}$$

The block diagram of the SAR ADC is shown in Figure 2.12. It consists of a sample and hold circuit (S/H), SAR, DAC, and a clocked comparator. N is the resolution and f_{ck} is the clock of the ADC. The clock generator generates the sampling clock (f_s), SAR and comparator clock (f_c), and the Reset clock (f_{rst}) using the ADC clock f_{ck} .

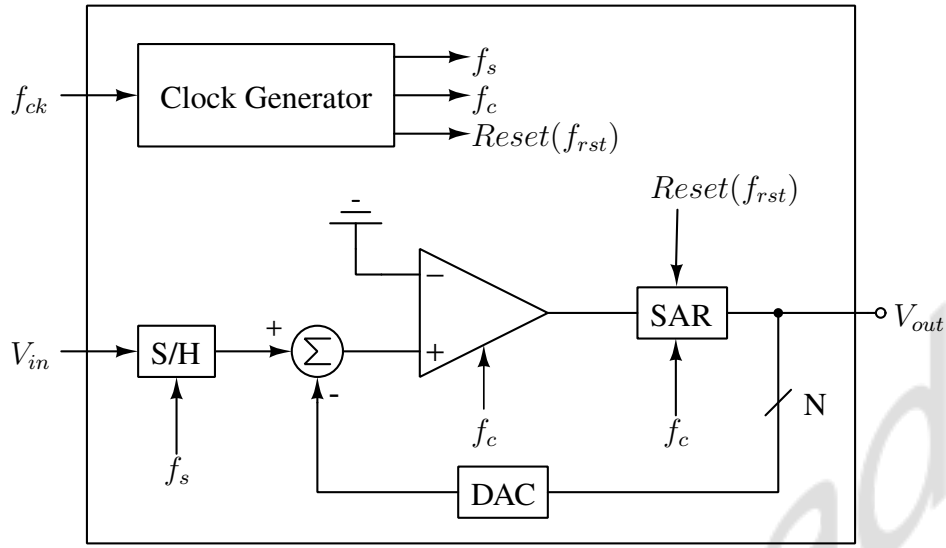


Figure 2.12: Block Diagram of SAR ADC [2]

Working Principle of SAR ADC

The SAR ADC determines the digital output by iteratively narrowing the search space across the available quantization levels (uses binary search algorithm). The conversion begins by setting the Most Significant bit (MSB) of the SAR to logic 1 and the remaining bits to logic 0. This digital output is converted to an analog signal using the charge sharing DAC in the feedback. The sampled input is evaluated against the DAC output through a dynamic comparator. If the sampled input is greater than DAC output, MSB will be retained as logic 1, and the next significant bit will be set to 1, and the remaining bits will be in the previous state. If the sampled input is smaller than DAC output, MSB bit will be reset to logic 0, and the next significant bit will be set to logic 1, and the remaining bits will be in the previous state. The digital output is again converted back to an analog signal and compared with the sampled input. A similar operation continues till the digital output converges to the required value.

Table 2.1 shows an example of binary search performed by a 4-bit SAR ADC. An N-bit SAR ADC takes N cycles of clock f_c to complete the analog-to-digital conversion. Hence, the signal should be sampled only once in every N cycles. The frequency of sampling clock f_s will be f_c/N .

When the signal is sampled at the Nyquist rate, the sampled data can lie anywhere in the dynamic range of the ADC. The difference between the consecutive samples can be equal to the dynamic range of the ADC in a few cases, as shown in Figure 2.13. It will be necessary to search through all quantisation levels if it is a Nyquist rate converter. Conventional SAR ADC

Comparator output	Output of ADC
	1000 (Initial state)
Logic 1	1100
Logic 0	1010
Logic 1	1011
Logic 1	1011 (Required output)

Table 2.1: Binary Search performed by a 4-bit SAR ADC

is the right architecture for such cases, as it searches through all the quantisation levels.

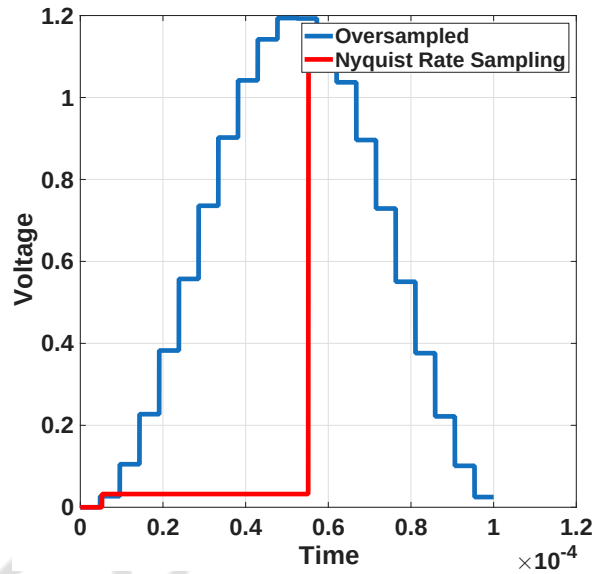


Figure 2.13: Sine wave sampled at Nyquist rate v Oversampled sine wave

If the input signal is oversampled, then the difference between the consecutive samples is minimal. For example, if a single tone signal with a frequency of 10 kHz and an amplitude of 1V ($1 \times \sin(2\pi 10^4 t)$) is sampled at 1 MHz, the peak difference between the consecutive samples is $\sin(2\pi 10^4 / 10^6) = 62$ mV. For an 8-bit ADC with a reference voltage of 1.8 V (step size = $1 \text{ LSB} = 1.8/2^8 = 7$ mV), 62 mV of variation in the consecutive samples will be equivalent to approximately 8 LSBs.

SAR ADC resets the output after analog-to-digital conversion of each sample. This means that MSB capacitors in Charge sharing DAC (CDAC) in the feedback will be set and reset in every cycle. This will charge and discharge all capacitors of the CDAC, which will consume power. For an over-sampled converter where sampled inputs do not vary significantly, it would

be unnecessary to perform reset operation after every conversion cycle. In the upcoming section, a modified version of SAR ADC is discussed, which aims to reduce the switching in the output of the ADC.

2.2.3 Least Significant Bit First SAR ADC (LSB-First SAR ADC)

In section 2.2.2, it was discussed that SAR ADC uses a binary search algorithm to converge into the required digital output. This section discusses a modified version of SAR ADC architecture titled “LSB-First SAR ADC” [7]. The difference between the two architectures is their search algorithm. An LSB-first SAR ADC does the following:

- The initial estimate for the search is derived from the preceding conversion.
- The iterations begin from the LSB side, setting or resetting bits starting from LSB, depending on the input. Hence, with each iteration, the resolution of the DAC reduces exponentially, i.e. its effective step size grows exponentially.
- This gradually builds up the DAC step size until the input voltage is represented with that resolution.
- From here, conventional binary search is used to quantise the input to the required resolution, i.e. the resolution of DAC increases exponentially in every iteration to locate the input. This phase is similar to the last few cycles of a conventional SAR ADC.

One may note the following key aspects of LSB-First SAR ADC algorithm:

- Unlike conventional/MSB-first SAR, LSB-First SAR ADC does not immediately swing the DAC by $V_{ref}/2$, which reduces large voltage transitions.
- The number of cycles taken for conversion is not constant. It can range from 2 to $2N$ cycles.

Working of LSB-First SAR ADC

LSB-First SAR ADC works in two phases: the LSB phase and the MSB phase. LSB phase starts with an initial guess, unlike conventional SAR ADC, which has a fixed starting point $\{V_{ref}/2\}$ for every conversion cycle. This architecture is designed for applications having an input signal

with less mean activity. So, the initial guess is taken to be the digital output of the previous sampled input. The initial guess is converted to an analog signal using the DAC in feedback and compared with the sampled input using a dynamic comparator. If the output of the comparator is logic 1, the first bit from the Least Significant bit (LSB) side, which is logic 0, is switched to logic 1 (vice versa if the comparator output is logic 0 initially). The digital output is converted to an analog signal and is compared sampled input again. This will continue until there is a transition in the comparator output. The LSB phase ends, and the MSB phase begins when there is a transition in the comparator output. An example of the LSB phase is shown in Table 2.2 in which the LSB phase ends after 3 cycles. MSB phase operation is similar to conventional SAR ADC operation. The difference is that it will not start from MSB. Instead, it will start from the bit next to the bit that was switched in the last cycle of the LSB phase. In the example below, 7th bit is the last changed bit in the LSB phase. So, the MSB phase starts with the 6th bit, which is highlighted in Table 2.2.

	Initial state of digital output of ADC:11010101101
Comparator output=1	After 1st cycle:11010101111
Comparator output=1	After 2nd cycle:11010111111
Comparator output=1	After 3rd cycle:11011111111
Comparator output=0	End of LSB phase and the beginning of MSB phase:11011111111

Table 2.2: LSB phase of LSB-First SAR ADC

Table 2.3 shows an example describing the working of the LSB-First SAR ADC. LSB-First SAR ADC, unlike SAR ADC, will not take a fixed number of cycles for analog-to-digital conversion. It will take 2 to 2N+1 cycles (N is the resolution of the ADC), depending upon the previous and current sampled input. With respect to Figure 2.12, the sampling clock frequency for

1. Conventional SAR ADC, $f_s = f_c/N$
2. LSB-First SAR ADC, $f_s = f_c/(2N + 1)$

As outlined in Chapter 1, conventional ADC architectures encounter performance limitations while targeting high resolutions. These are mainly due to the offset in the comparator and non-linearities which impose boundaries and make it is near impossible to achieve a resolution

Comparator output	Digital output of ADC	ADC operating phase
	1101000110	Initial state
Logic 1	1101000111	LSB phase
Logic 1	1101001111	LSB phase
Logic 1	1101011111	LSB phase
Logic 0(First transition)	1101010111	MSB phase
Logic 1	1101011011	MSB phase
Logic 0	1101011001	MSB phase
Logic 1	1101011010	MSB phase
Logic 1	1101011011	MSB phase
End of conversion	1101011011	Required output

Table 2.3: Example table showing the working of the LSB-First SAR ADC [7]

beyond 12 to 14 bits [1]. These motivated to explore alternate strategies to achieve the required performance by overcoming the imperfections in analog systems.

Oversampling Analog-to-Digital Converters

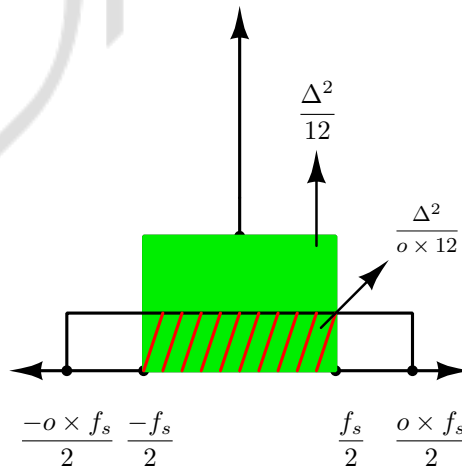


Figure 2.14: Noise-Power spectral density for a Nyquist-rate and Oversampling ADC with noise power of oversampled ADC being o times less than that of Nyquist-rate ADC where o is the OSR

In analog-to-digital converters, the sampling frequency plays a critical role in determining how quantization noise is distributed across the frequency spectrum. For an ADC with a

fixed resolution, the total quantization noise power remains essentially constant, as it is determined by the quantization step size. However, increasing the sampling frequency expands the Nyquist bandwidth over which this noise power is spread. As a result, the noise spectral density is reduced within the signal band of interest, leading to a lower in-band noise floor. This redistribution of noise effectively improves the signal-to-noise ratio (SNR) within the desired bandwidth, even though the total noise power remains unchanged. The impact of this noise spreading with higher sampling frequencies is illustrated in Figure 2.14. The SNR of the oversampling converter is given by

$$SNR = 6.02N + 1.76 + 10\log(f_s/2f_{in}) \quad (2.9)$$

2.2.4 Delta Sigma Modulator (DSM)

Previous sections discussed the Nyquist rate converters and their imperfections, which limit the maximum achievable resolution. This led to a platform for the introduction of DSM, which exploits oversampling and noise-shaping to achieve higher resolutions while relaxing hardware requirements. The working principle of DSM is detailed in this section.

In a conventional ADC architecture, the digital output can be expressed as

$$V_{out} = V_{in} + Q_e, \quad (2.10)$$

where Q_e denotes the quantization error, the magnitude of which is determined by the resolution of the ADC. Improving the noise efficiency of an ADC, therefore, requires minimizing this quantization error.

One effective approach to suppress Q_e is the introduction of a negative feedback mechanism. Through feedback, the quantization error can be driven toward zero, thereby forcing the output V_{out} to closely track the input V_{in} , as illustrated in Figure 2.15. Achieving such behavior requires the signal transfer function (STF) and the noise transfer function (NTF) to be different. In conventional ADCs, both transfer functions are unity, as implied by Equation 2.10, resulting in no preferential treatment of signal and noise.

As shown in Figure 2.15, the architecture incorporates an Anti Aliasing Filter (AAF) at the input and a reconstruction filter at the output of the DAC to suppress out-of-band noise components. Consequently, only the low-frequency portion of the quantization noise overlaps with the signal band, where it can be effectively corrected by the negative feedback loop. Since

the AAF and the DAC output filter exhibit identical characteristics, the system in Figure 2.15 can be equivalently represented by the simplified model shown in Figure 2.16.

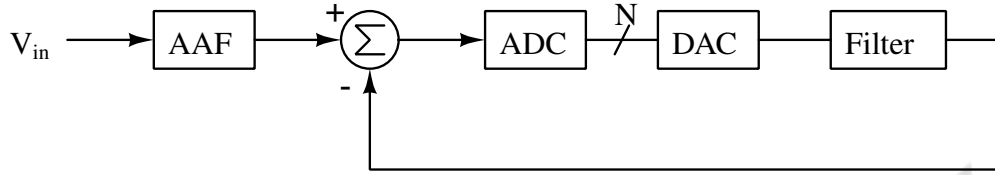


Figure 2.15: ADC architecture with a negative feedback to reduce the quantization noise

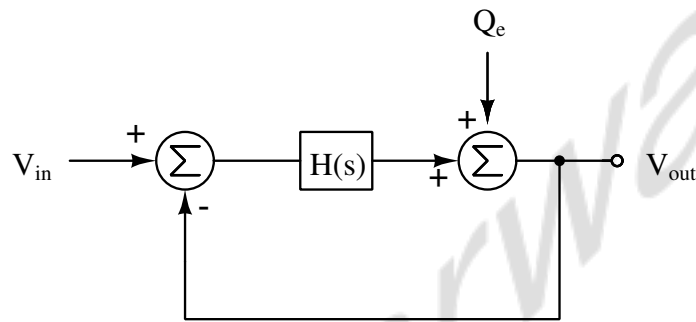


Figure 2.16: Noise model of DSM

With respect to the noise model in Figure 2.16, the output of DSM is given by

$$V_{out} = V_{in}(STF) + Q_e(NTF) \quad (2.11)$$

where,

1. STF - signal transfer function.
2. NTF - noise transfer function.

The STF and NTF are equal to 1 in the case of Nyquist-rate converters. STF and NTF of the DSM are given by

$$STF = \frac{H(s)}{H(s) + 1} \text{ and } NTF = \frac{1}{H(s) + 1} \quad (2.12)$$

To perform noise attenuation at lower frequencies, as shown in Figure 2.17, NTF should have a high-pass filter response, which means that $H(s)$ will be having low-pass filter response. So, $H(s)$ will be an integrator as shown in Figure 2.18 [4].

Continuous-Time Delta-Sigma Modulator is an oversampling converter consisting of an integrator and a 1-bit ADC in the form of a comparator, as shown in Figure 2.18. In the DSM,

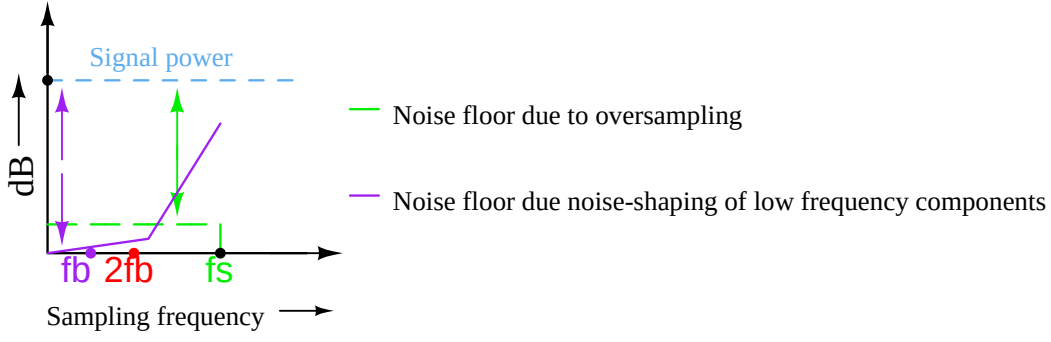


Figure 2.17: Noise Floor of ADC

the input going into the integrator will be an error signal given by $e(n) = V_{in}(n) - V_{out}(n-1)$. Integrator accumulates the error $e(t)$ over time, which is fed as input to the 1-bit quantizer. The comparator does not just see the raw input, unlike a simple 1-bit quantiser (or comparator) without any feedback system. It sees an accumulated error due to which DSM generates a pulse density modulated output. An oversampled ADC will have less noise power inside the

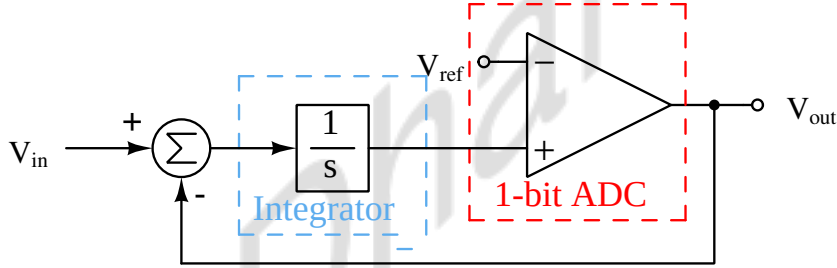


Figure 2.18: Block diagram of Delta-Sigma Modulator [5]

signal band as shown in Figure 2.17. This is because an oversampled ADC has more samples compared to a Nyquist rate ADC and is capable of giving a better approximation of the original signal after reconstruction. DSM is an oversampling converter. The output equation of DSM is given by

$$V_{out} = \frac{V_{in}}{1 - e^{-T}z^{-T}} + Q_e \left(1 - \frac{1}{1 - e^{-T}z^{-T}} \right) \quad (2.13)$$

where Q_e is quantisation error as shown in Figure 2.16. So, the noise transfer function is $s/(s+1)$ (transfer function of ideal integrator is $1/s$), it is evident that its AC response resembles that of a high-pass filter. Due to this, lower frequency components will be attenuated, leading to the reduction in the noise power in the lower frequency band as shown in Figure 2.17. This boosts the SNR. DSM can achieve an SNR equivalent to a 16-bit ADC using a 1-bit ADC by attenuating low-frequency noise components through noise-shaping. SNR of 1-bit DSM fed

with a signal of frequency f_{in} sampled at frequency f_{ck} is given by

$$SNR = 6.02 + 1.76 + 30\log(f_{ck}/2f_{in}) - 10\log(\pi^2/3) \quad (2.14)$$

It can be observed from Equation 2.14 and 2.14 that SNR is improved by $20\log(f_{ck}/2f_{in}) - 10\log(\pi^2/3)$. As discussed earlier, the DSM is capable of achieving an SNR equivalent to a 16-bit ADC. To achieve this, $f_{ck}/2f_{in}$ should exceed 1000, which will compromise the input signal bandwidth as f_{ck} is limited by the technology.

To resolve this issue, a hybrid architecture titled noise-shaping ADC was introduced, which consists of an N-bit ADC used as a quantiser for the DSM, as shown in Figure 2.19. SNR of multibit DSM fed with a signal of frequency f_{in} sampled at frequency f_{ck} is given by

$$SNR = 6.02N + 1.76 + 30\log(f_{ck}/2f_{in}) - 10\log(\pi^2/3) \quad (2.15)$$

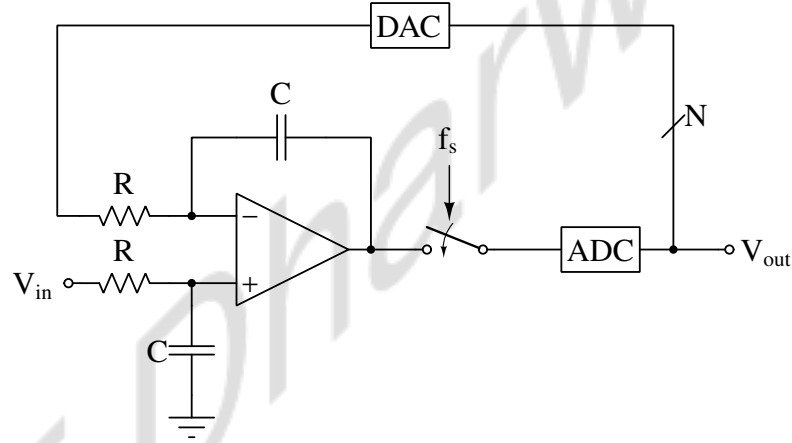


Figure 2.19: N-bit Continuous-Time Noise-Shaping ADC

Now, a 6-bit ADC with $f_{ck}/2f_{in}$ equal to 100 can achieve an SNR equivalent to a 16-bit ADC. This will improve the input signal bandwidth by a factor of 10. Quantiser can be chosen to be Flash ADC or SAR ADC. Noise-Shaping ADC is primarily designed for applications requiring moderate speed and high noise immunity. Using Flash ADC as a quantiser would result in unnecessary power consumption and hardware complexity. A SAR ADC, on the other hand, provides an ideal balance of speed, power, and resolution, making it a highly suitable quantiser choice for delta-sigma modulator (DSM) designs. The main benefits of integrating SAR ADC as a quantiser in DSM include:

1. Power-efficient architecture
2. The maximum achievable resolution is more in this architecture

2.3 Problem Formulation

SNR of ADC is given by

$$SNR = 6.02N + 1.76 + 10\log\left(\frac{f_s}{2f_{in}}\right) \quad (2.16)$$

where,

1. N is the resolution of ADC.
2. f_{in} is input signal bandwidth.
3. f_s is sampling frequency.

Equation 2.16 clearly indicates that the signal-to-noise ratio (SNR) improves linearly with the resolution of the ADC. Specifically, each additional bit of resolution reduces the quantization step size, thereby lowering the quantization noise power and resulting in a corresponding increase in SNR. In addition, the equation shows that SNR is directly related to the sampling frequency: increasing the sampling rate spreads the quantization noise over a wider frequency range, which reduces the noise power within the signal band. Conversely, for a given sampling frequency, an increase in the input signal bandwidth leads to a higher amount of in-band noise, thereby degrading the SNR. Thus, higher resolution and higher sampling frequency improve SNR, while wider input bandwidth adversely affects it.

2.3.1 SNR of SAR ADC

For SAR ADC, sampling frequency $f_s = f_c/k$, where f_c is SAR clock and k is the number of cycles taken for analog-to-digital conversion. SNR for SAR ADC is given by

$$SNR = 6.02N + 1.76 + 10\log\left(\frac{f_c}{2f_{in}(k)}\right) \quad (2.17)$$

From the previous discussion regarding the SNR, we can conclude that it indicates the quality of the signal. Higher SNR indicates that signal quality is better. In Equation 2.17 point of view, SNR of ADC can be improved by doing the following.

1. Increase the resolution of the ADC, N .
2. Increase the SAR controller clock frequency, f_c .

3. Decrease input signal bandwidth, f_{in} .
4. Decrease the number of cycles taken for analog-to-digital conversion, k .

SAR controller clock frequency, f_c , is limited by technology, and decreasing f_{in} will reduce the input signal bandwidth of the ADC. Increasing the resolution of the ADC (N) can also increase the SNR. But, it is limited by the comparator offset and CDAC non-linearity. For a conventional SAR ADC, $k=N$. To improve the SNR of the ADC without affecting the bandwidth or by increasing the ADC's resolution, it is necessary to decrease the number of cycles taken for analog-to-digital conversion, k . This will increase the speed of ADC(decreases latency), which will be useful in improving SNR. This work discusses an approach to achieve the same, whose details and implementation are discussed in Chapter 3.

2.3.2 SNR of Noise-Shaping SAR ADC

SNR of 1st-order Noise-Shaping SAR ADC with SAR clock frequency, f_c and number of cycles per conversion k is given by

$$SNR = 6.02N + 1.76 + 30\log\left(\frac{f_c}{2f_{in}(k)}\right) - 10\log\left(\frac{\pi^2}{3}\right) \quad (2.18)$$

Reducing the number of clock cycles required per conversion, denoted by k , effectively increases the sampling rate of the ADC for a fixed clock frequency. Since each conversion completes faster, more samples are acquired within the same time interval, leading to a higher effective sampling frequency. This increase in sampling rate results in a higher oversampling ratio (OSR) for a given signal bandwidth.

To further enhance the SNR, the ADC is incorporated as a multibit quantizer within a delta-sigma ($\Delta\Sigma$) modulation loop. In this configuration, the loop employs noise shaping, which actively pushes the quantization noise toward higher frequencies outside the signal band. The combination of increased sampling rate (due to reduced k), oversampling, and noise shaping in the $\Delta\Sigma$ loop significantly suppresses in-band noise. This leads to a significantly larger improvement in SNR compared to what can be achieved through oversampling alone, as captured by Equation 2.18. Consequently, the proposed architecture becomes highly suitable for high-resolution data conversion.

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Chapter 3

Non-Uniform Sampling ADC: Methodology and Implementation techniques

In Section 2.3, SNR of ADC and possible ways to improve it was discussed. SNR of ADC in SAR ADC point of view was also discussed. In this Chapter, a methodology to improve the SNR of SAR ADC and system level design of it will be discussed.

3.1 Working principle of Non-Uniform sampling ADC

One of the notable differences between SAR ADC and LSB-First SAR ADC is the initial state of the output of the ADC for each conversion cycle. For every conversion cycle, SAR begins from an initial state. It can be observed from Figure 3.1 that the window between the initial and final state for the 2^{nd} input sample is smaller in the case of LSB-First SAR ADC. This way LSB-First SAR ADC can achieve faster conversion. The original authors of this technique used a fixed sampling rate corresponding to one sample every $2N+1$ cycles [7], i.e. every sampled input is allotted a dedicated number of conversion cycles (i.e., $2N+1$) as discussed in Section 2.2.3. This means if the conversion completes in 2 cycles, the ADC will stay idle for the remaining $2N-1$ cycles, leading to inefficient use of available conversion cycles.

In this mode, the LSB-First SAR ADC is fed with an input sample once in every $2N+1$ cycles. Alternatively, if we feed an input sample at the end of each analog-to-digital conversion by eliminating the idle time of the ADC, we can effectively feed more input samples into the

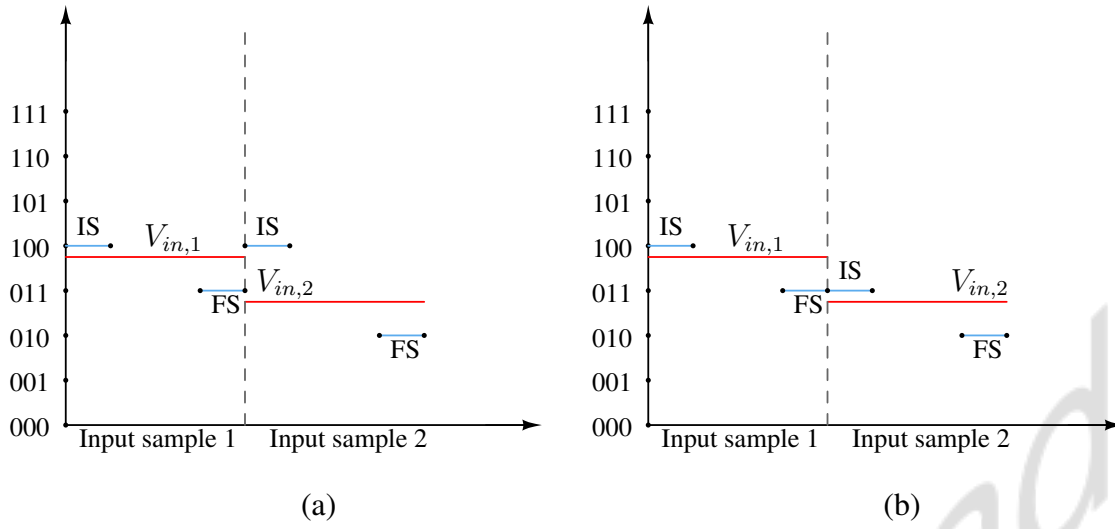


Figure 3.1: Examples showing Initial state (IS) and Final state (FS) of (a) SAR ADC and (b) LSB-First SAR ADC for different sampled inputs, $V_{in,1}$ and $V_{in,2}$

ADC in a given period. This effectively increases the achievable sampling rate without varying the SAR clock frequency (f_c). We are effectively reducing the number of clock cycles of the SAR clock f_c consumed for analog-to-digital conversion, to improve the throughput by improving the sampling frequency as shown in Figure 3.2. This is particularly useful in oversampled converters as will be discussed later in this work.

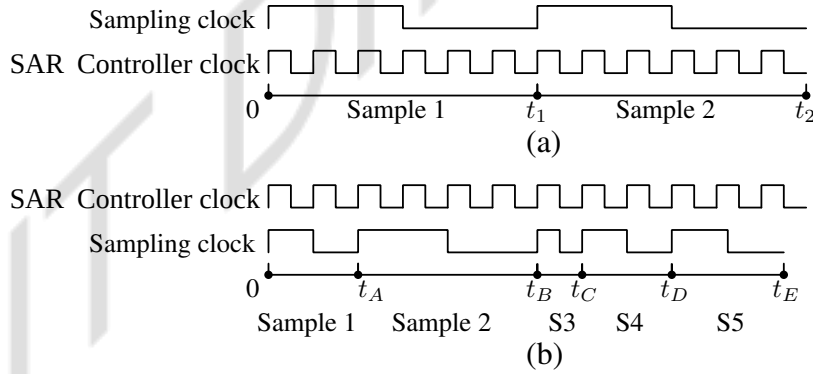


Figure 3.2: Example sampling clock for (a) SAR ADC and (b) Implemented ADC architecture

3.2 Design of Non-Uniform Sampling ADC

In section 2.2.3, it was discussed that the LSB-First SAR ADC works in two phases: the LSB and the MSB phase. These phases are separated by the transition in the comparator output. It is necessary to have a logic to detect the first transition in the comparator output, which will

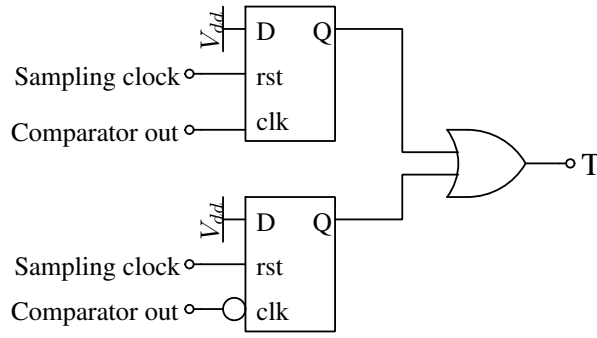


Figure 3.3: Transition detector

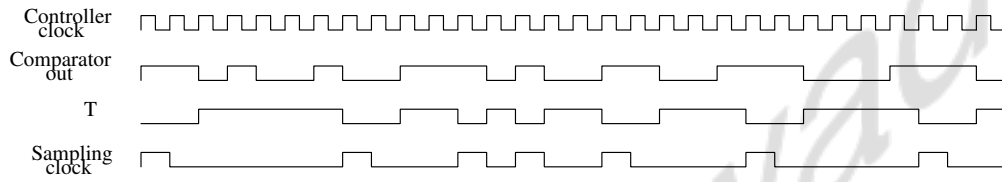


Figure 3.4: Example waveform showing working of Transition Detector

be useful in building a SAR logic to operate in 2 different phases. Digital logic in Figure 3.3 detects the transition in the comparator and output of the logic, T will switch to V_{dd} and will remain in the state for the rest of the conversion cycle, as shown in Figure 3.4. T will remain V_{dd} state for the rest of the time irrespective of any transitions occurred. Hence, it is necessary to reset the logic at the end of every conversion cycle. So, the reset in this logic is controlled by the sampling clock(clks).

In the LSB phase, it is necessary to detect the 1st bit from the LSB side, which is logic 1 or 0 depending on the comparator output, which is detailed in Section 2.2.3. To detect the same, Priority Encoder (PE) and decoder are used as shown in Figure 3.5. Input to this will be the register array 2, which holds the output of SAR. Register array 1 holds the decoder data in the LSB phase. Using the data stored in register array 1 set/reset operation is carried out on output registers (Register array 2) using the digital logic shown in Figure 3.6.

In the MSB phase, the SAR operation in the LSB-First SAR ADC is similar to that of a conventional SAR ADC. SAR of the LSB-First SAR ADC should resemble conventional SAR logic shown in Figure 3.7. For that to happen PE should be disabled, register array 1 should behave like shift registers, and registers in register array 2 should be clocked, controlled by the shift register. PE is disabled by disabling the MUX connected to its input, which is controlled by the transition detector. Similarly, select lines of the MUX in register array 1 are controlled by a transition detector, using which they can be made to behave like shift registers. Similarly,

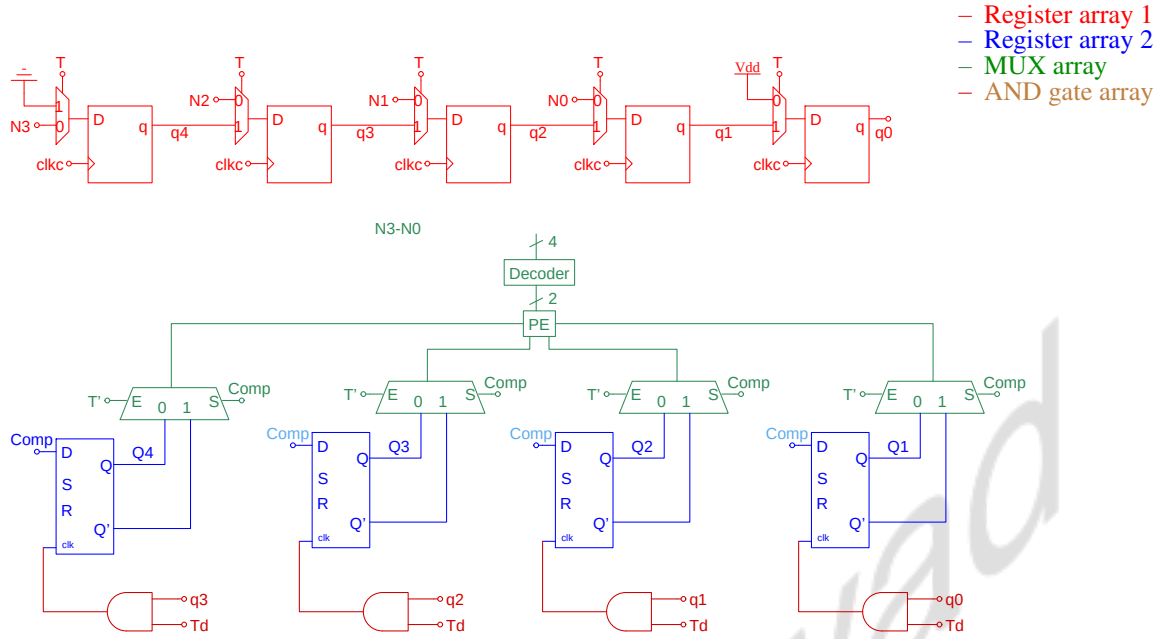


Figure 3.5: 4-bit LSB-First Successive Approximation Register implementation for Variable-rate SAR ADC (Comp denotes Comparator output and clk_c denotes LSB-First SAR controller clock)

the clocks of register-array-2 are controlled.

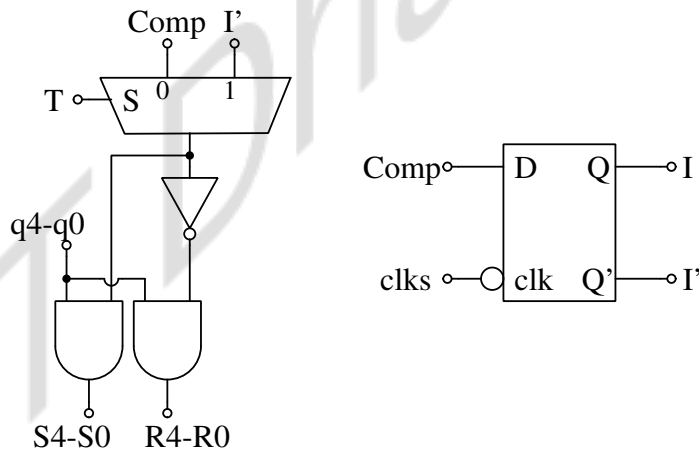


Figure 3.6: Digital logic to control set and reset operations of output registers in LSB-First SAR (Comp denotes Comparator output and clks denotes sampling clock of the ADC)

In Figure 3.8, it can be observed that register q will be zero at the end of each conversion. This state of register can be detected to trigger the sampling clock to sample the new input into the ADC. This way, idle time in the LSB-First SAR ADC is eliminated, and analog-to-digital conversion becomes quicker. Since the idle time of the ADC is eliminated, the period of each sampling clock cycle will be f_c/k where, $k \in 2, 3, 4, \dots, 2N + 1$ (N is the resolution of ADC).

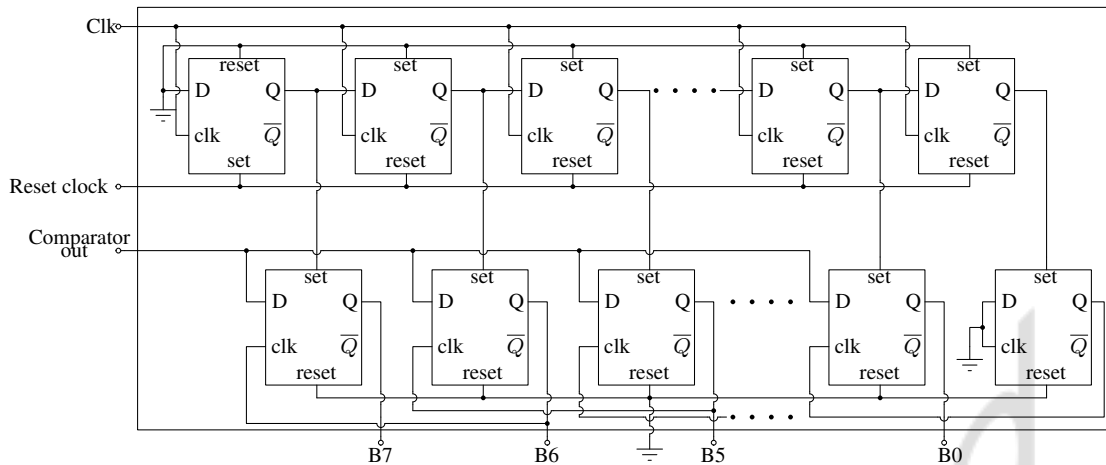


Figure 3.7: 8-bit conventional successive approximation register [2]

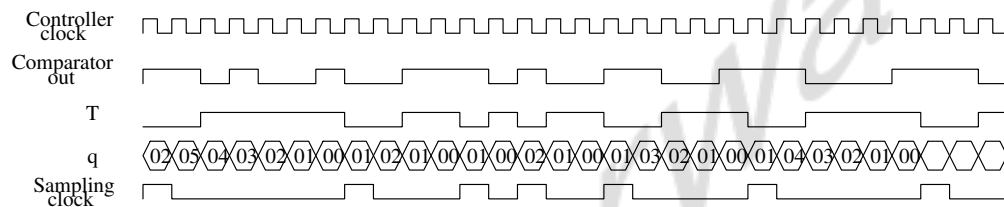


Figure 3.8: Example waveform showing generation of sampling clock for Variable-rate SAR ADC

This means that the period of each sampling clock is not fixed. Hence, this ADC architecture is titled “**Non-Uniform Sampling ADC**”. The block diagram of the Non-Uniform Sampling ADC is shown in Figure 3.9.

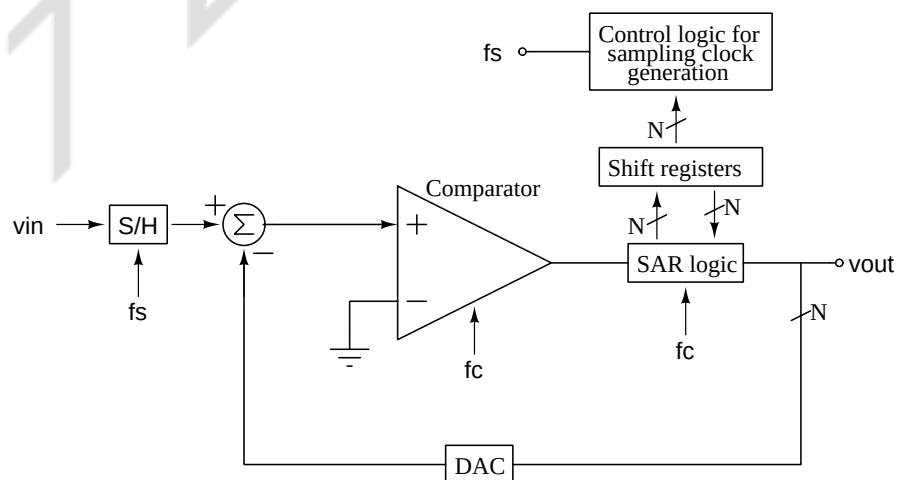


Figure 3.9: Block diagram of Non-Uniform Sampling SAR ADC

Sequence of operations:

1. Input signal is sampled into the ADC while the SAR logic resets itself to an initial guess. The initial guess will be the output of the previous sample as input is not a busy signal.
2. Output is converted to analog and is compared with the sampled input. Based on the comparator generates the control signal which guides the SAR to converge into required digital output. SAR consists of LSB-First search algorithm embedded in it discussed in Chapter 2.
3. The shift register keeps a track bit update operations in the SAR. Control logic used for sampling clock generation tracks the bit update operation to trigger the sampling clock at the end of each conversion.
4. Similar procedure continues, to converge the new sampled input into required digital output.

3.3 Observations

An 8-bit architecture was modeled and tested with a controller frequency $f_c = 10\text{MHz}$ across different input signal bandwidths. Number of cycles per conversion can vary from 2 to 16 cycles, and respective data is displayed in the Figures 3.10, 3.11, 3.12, 3.13, and 3.14. The following are observations:

1. System is Linear-Time Variant (LTV)

- (a) ADC may take any number of cycles, varying from 2 to $2N+1$. So, the next input sampled into the ADC and the corresponding digital data will be dependent on the speed of conversion or the number of cycles taken by the previous sampled input for analog-to-digital conversion. Sample spaces are not uniform. Conventional SAR ADC has uniform sample spaces.
 - (b) Non-uniform sample spaces where each space is guided by the previous input sample makes the architecture a Time-Variant system.
2. Frequency of number of samples taking more than 8 cycles is less for smaller bandwidths. Input will be busy in a higher bandwidth signal, impacting the average cycle consumed per conversion proportionally as shown in Table 3.1.

3. In a conventional SAR ADC, the bandwidth of a signal is compromised to improve or increase the number of samples per cycle. The variable OSR architecture increases the number of samples sampled into the ADC for the same input signal bandwidth.

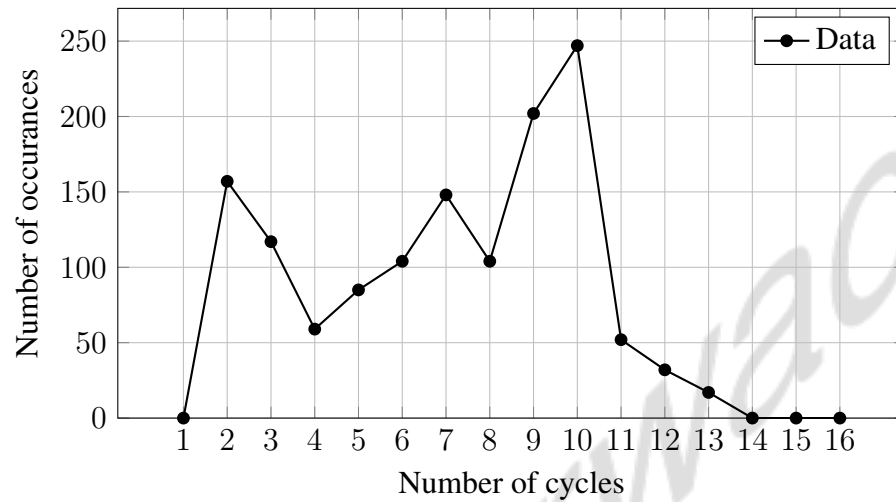


Figure 3.10: Distribution total number of cycles per conversion taken by Variable-rate SAR ADC over a period for input signal frequency=25000 Hz and ADC clock frequency=10 MHz

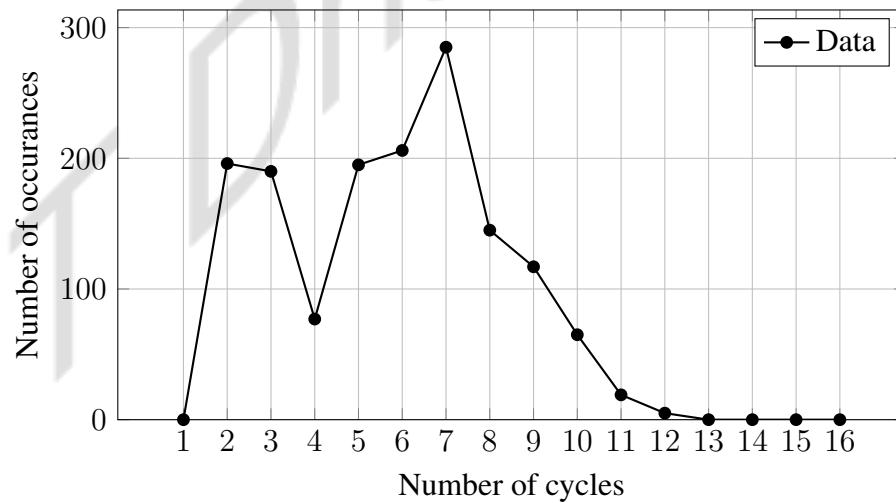


Figure 3.11: Distribution total number of cycles per conversion taken by Variable-rate SAR ADC over a period for input signal frequency=12500 Hz and ADC clock frequency=10 MHz

Variable-rate LSB-First SAR ADC has been tested for different input signals, and the average cycles per conversion (ACPC) are compared with the conventional SAR ADC and listed in the Table 3.1.

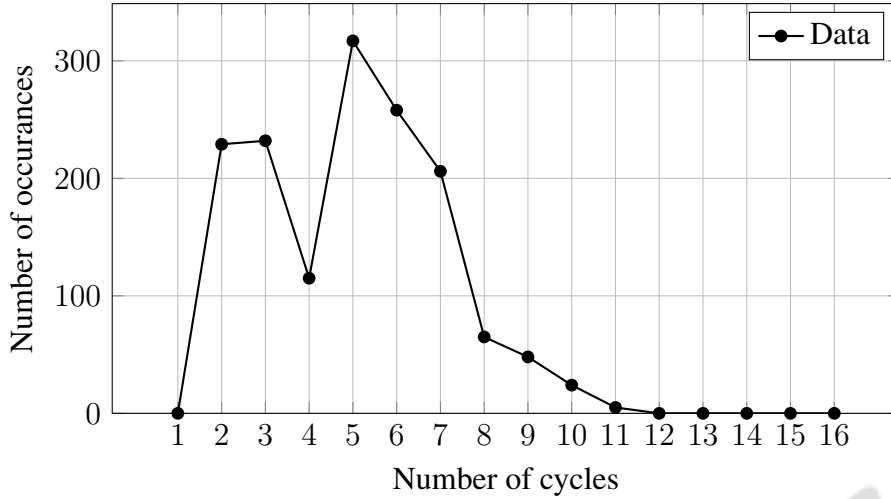


Figure 3.12: Distribution total number of cycles per conversion taken by Variable-rate SAR ADC over a period for input signal frequency=7812.5 Hz and ADC clock frequency=10 MHz

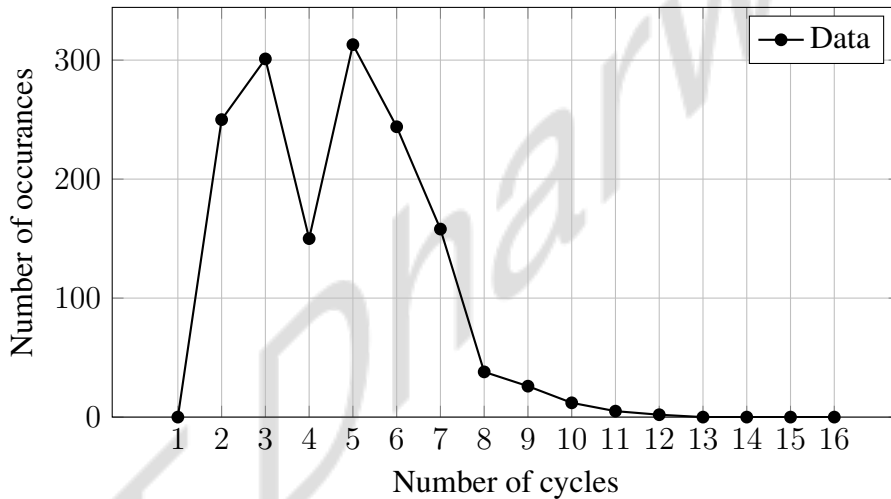


Figure 3.13: Distribution total number of cycles per conversion taken by Variable-rate SAR ADC over a period for input signal frequency=6250 Hz and ADC clock frequency=10 MHz

3.4 Effective OSR improvement and its impact on SNR of ADC

Figure 3.1 compares the working of conventional SAR ADC and LSB-First SAR ADC, while Figure 3.15 compares the working of LSB-First SAR ADC and the variable-rate SAR ADC. The ADC involves asynchronous termination of the conversion cycle, once the conversion is completed. This is done by eliminating the idle time of ADC. The difference in the consecutive samples decreases further, which is evident in Figure 3.15. This way, effective OSR increases.

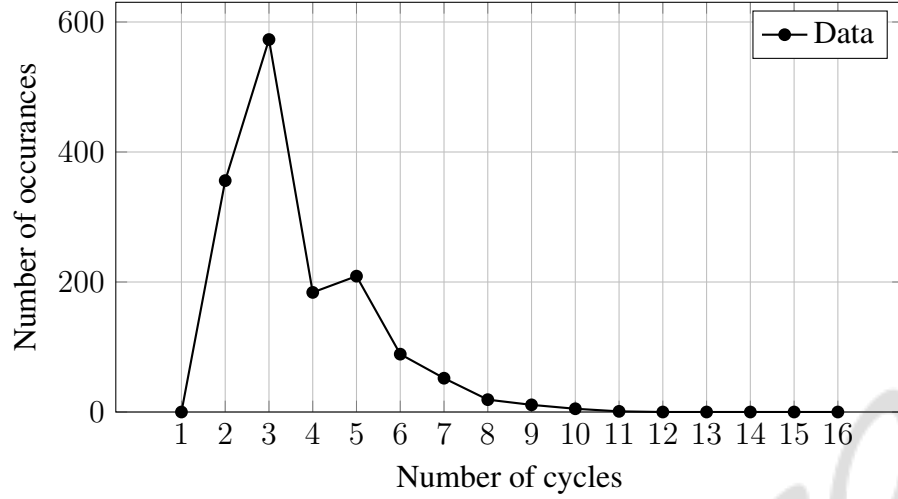


Figure 3.14: Distribution total number of cycles per conversion taken by Variable-rate SAR ADC over a period for input signal frequency=2500 Hz and ADC clock frequency=10 MHz

Test bench	INPUT SIGNALS		SAR ADC	LSB-FIRST SAR ADC
Serial No.	f_c (in Hz)	f_{in} (in Hz)	ACPC	ACPC
TB1	10M	25000	8	6.55
TB2	10M	12500	8	5.3
TB3	10M	7812.5	8	4.49
TB4	10M	6250	8	4.12
TB5	10M	3125	8	3.5
TB6	10M	2500	8	3.2

Table 3.1: Observations for different test benches for the implemented Variable-rate SAR ADC

The increase in effective OSR implies that the sampling frequency is increased, as well as the number of data points sampled into the ADC. More samples of a signal will give a closer approximation of the reconstructed signal compared to the original signal. Due to this, the noise floor of the ADC will decrease, as shown in Figure 3.16.

SNR of ADC is given by

$$SNR = 6.02N + 1.76 + 10 \log_{10} \left(\frac{f_s}{2f_{in}} \right)$$

where,

$$f_s = \frac{f_c}{k} \text{ for SAR ADC.}$$

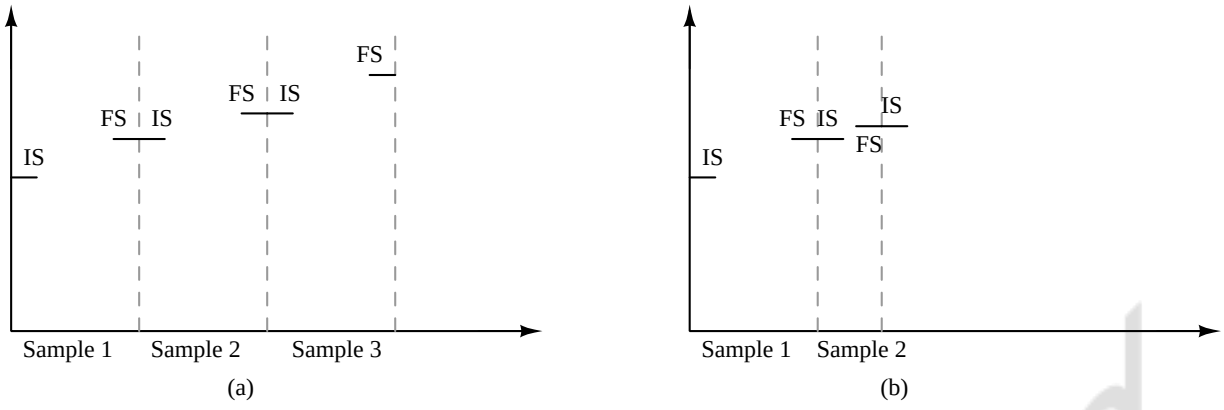


Figure 3.15: Examples showing Initial state (IS) and Final state (FS) of (a) LSB-First SAR ADC and (b) Variable-rate SAR ADC for different samples, Sample 1, 2, 3

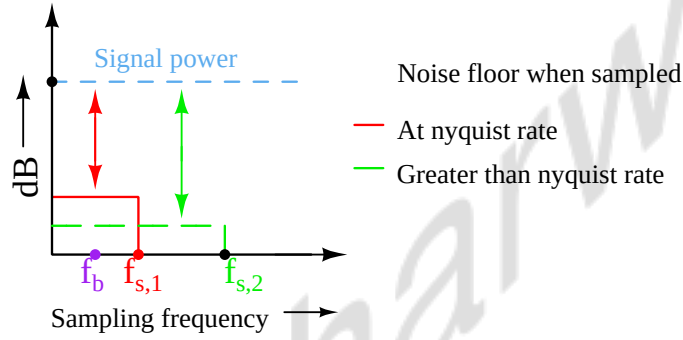


Figure 3.16: Noise Floor of ADC

In this work, SAR controller clock f_c is kept constant, and the number of cycles taken for analog-to-digital conversion (k) is reduced to improve the latency of the ADC. This way, without affecting any parameters effective sampling frequency is increased, thereby improving the SNR.

3.5 Results

3.5.1 SNR and ENOB

- Frequency of the input signal is 5KHz
- Frequency of the controller clock signal is 16MHz.

An 8-bit Conventional SAR ADC takes 8 cycles per conversion irrespective of the input signal. The implemented 8-bit variable-rate LSB-First SAR ADC architecture takes 3.5 cycles

per conversion for the above specified test signals, leading to an improvement in the effective OSR. This effective improvement in OSR will improve the SNR by reducing the in-band noise, which can be observed in Figure 3.17.

Improvement in the SNR of the Variable-rate ADC compared to the conventional SAR ADC for the above test signals is given by 3 dB, which is equivalent to 0.5 Effective Number of Bits (ENOB).

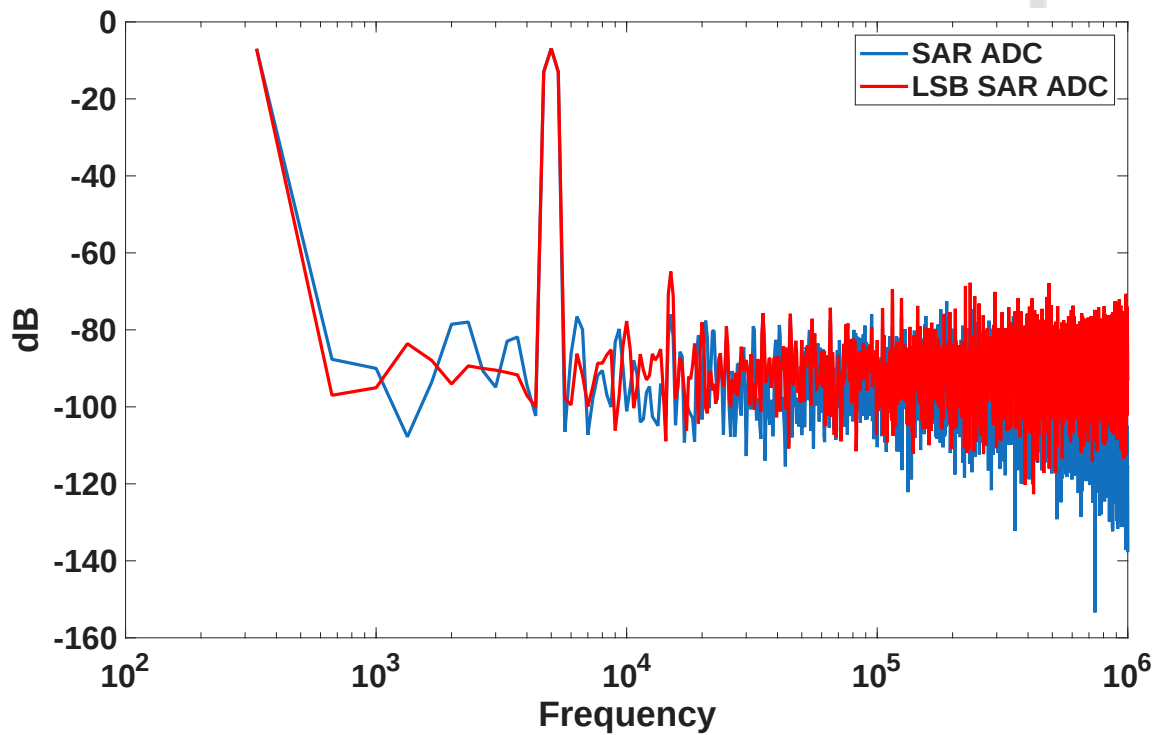


Figure 3.17: Spectrum of Conventional vs Variable-rate SAR ADC

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Chapter 4

Variable-Rate Delta-Sigma ($\Sigma\Delta$) ADC: Analysis and Design

In Chapter 2, working principles of DSM and how it exploits oversampling and noise-shaping to reduce the noise power at lower frequencies were discussed. They are designed for low-speed applications that require high noise immunity. To extend the high noise immunity advantages to moderate speed applications, high-resolution ADCs are used as quantizers in the $\Delta\Sigma$ architecture, making it a hybrid architecture that combines the advantages of both architectures. Usually, SAR ADC is used as a multibit quantizer for DSM, which makes the architecture power-efficient, highly immune to noise, and has moderate speed. In Chapter 2.1.4, a novel ADC architecture was introduced, which uses Non-Uniform Sampling techniques to improve the speed of the ADC, thereby improving the SNR.

This Chapter discusses the design and implementation of Continuous-Time DSM using the Non-Uniform Sampling ADC implemented in Chapter 3 as a quantiser. By adopting this approach, the noise power in the lower-frequency region is further suppressed, which was earlier reduced using the Non-Uniform sampling techniques.

4.1 1st order $\Delta\Sigma$ ADC

The block diagram 1st order noise shaping ADC is shown in Figure 4.1. The output is given by

$$V_{\text{out}} = STF(z)V_{\text{in}} + NTF(z)Q_e$$

where,

Q_e is quantization noise

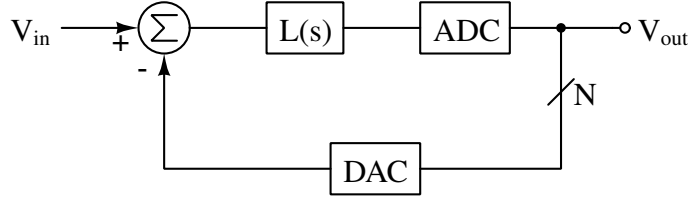


Figure 4.1: 1st order noise shaping ADC

4.1.1 Design of loop filter L(s)

An active filter is used to implement the loop filter shown in Figure 4.2. The transfer function of the loop filter L(s) will be

$$L(s) = \frac{-1}{sRC}$$

The signal and Noise Transfer Function for the respective loop filter will be

$$STF(z) = \frac{1}{RC(1 - e^{-T/RC} z^{-T})} \text{ and } NTF(z) = \left(1 - \frac{1}{RC(1 - e^{-T/RC} z^{-T})}\right) \quad (4.1)$$

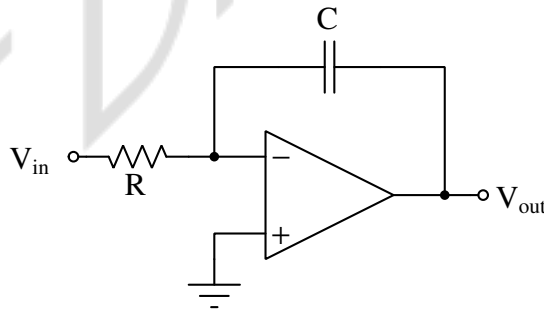


Figure 4.2: Integrator design for DSM loop filter

The above transfer functions are true if the gain of the amplifier is considered to be ∞ . For an amplifier with finite gain A, equations are rewritten as

$$L(s) = \frac{A}{1 + sRC(1 + A)}$$

$$NTF = \frac{1}{1 + A} \frac{1 + sRC(1 + A)}{1 + sRC}$$

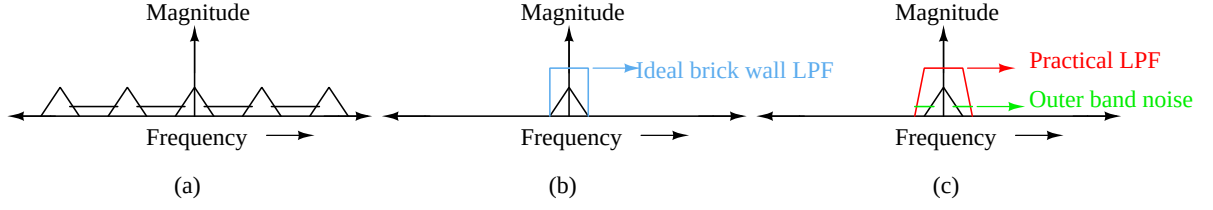


Figure 4.3: Spectrum of (a) ADC output, (b) Reconstructed signal using ideal filter, (c) Reconstructed signal using practical filter

Due to this finite gain of the amplifier, noise at lower frequency bands is not fully attenuated.

The choice of the noise shaping band plays a crucial role. This is because a reconstruction filter cannot have brick wall filter characteristics, due to which the signal will contain a few outer and frequency components after reconstruction, as shown in Figure 4.3. Proper design of the integrator leads to noise-shaping in DSM, which will attenuate the noise components at lower frequency bands. Rolloff in the practical reconstruction filter will not cause a major impact on the signal quality, as noise attenuation will also cover a few frequency components outside the input signal band, relaxing the filter requirements.

It is evident from Equation 4.1 that NTF has a zero located at the origin and a pole at $\frac{1}{RC}$. For proper noise shaping, RC is chosen as follows.

$$\frac{1}{RC} = \frac{f_s}{2}$$

The effect of RC product variation on the bandwidth of the loop filter $L(s)$ and NTF can be observed in Figure 4.4 and 4.5, respectively. Low RC will introduce the highest in-band attenuation, due to which the SNR will get better. But, this comes at a cost. It can be observed in Figure 4.5 that the gain at higher frequencies is increased. This will overload the quantiser, resulting in signal-dependent instability. The instability can be observed in Figure 4.6. This reduces the maximum amplitude of the input signal. The maximum amplitude of the input signal beyond which the DSM will be unstable, producing bounded internal states and a finite output without entering overload or limit-cycle behavior. It is the maximum amplitude of the input sinusoid that will not cause integrator saturation or quantizer overload. Hence, the nominal value of RC is chosen, i.e., $\frac{1}{RC} = \frac{f_s}{2}$.

Variable-Rate 1st-order Noise-Shaping SAR ADC is implemented and tested as shown in Figure 4.7. Considering k_{avg} to be the average number of cycles per conversion, the RC product

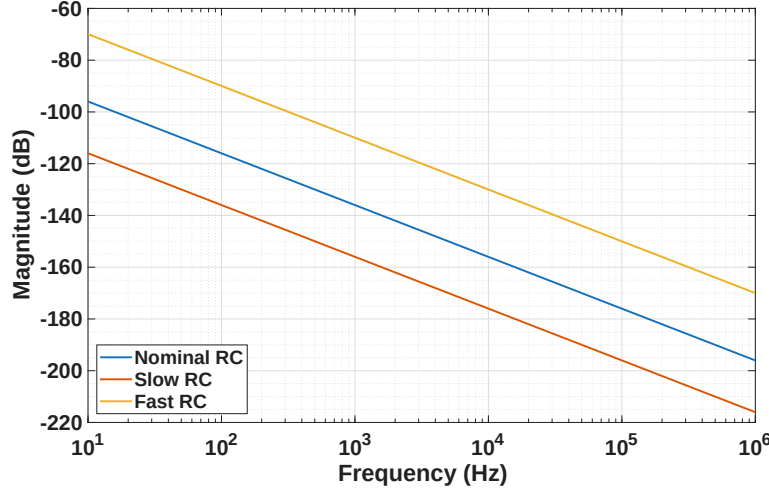


Figure 4.4: Magnitude of the loop-filter transfer function for the different RC products

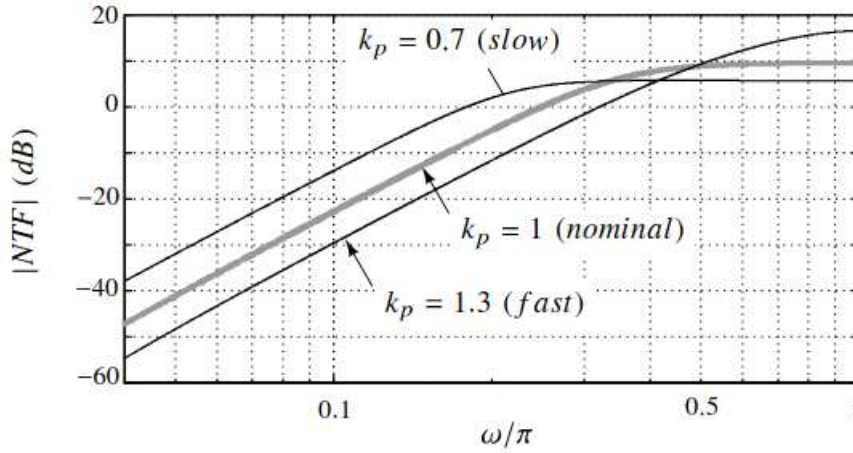


Figure 4.5: Magnitude of the noise transfer function for the different RC products [5]

for the design of the integrator is given by

$$\frac{1}{RC} = \frac{f_c}{2 \times k_{\text{avg}}} \quad (4.2)$$

where f_c is the SAR controller clock. In a continuous-time $\Delta\Sigma$ ADC, the input signal is processed in the continuous-time domain and is directly fed to the integrator, with sampling occurring only at the integrator output. Since the integrator exhibits low-pass filtering characteristics, high-frequency noise components in the input are significantly attenuated before sampling. Therefore, the use of a separate AAF to remove high-frequency noise at the input is unnecessary.

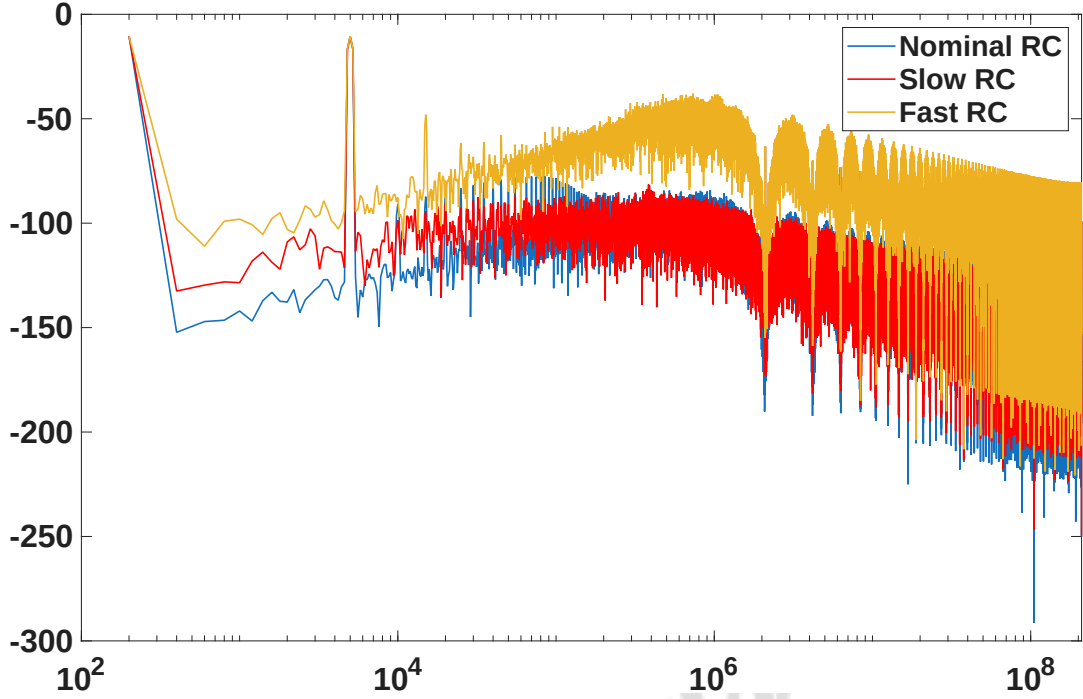


Figure 4.6: Spectrum of DSM for different RC products

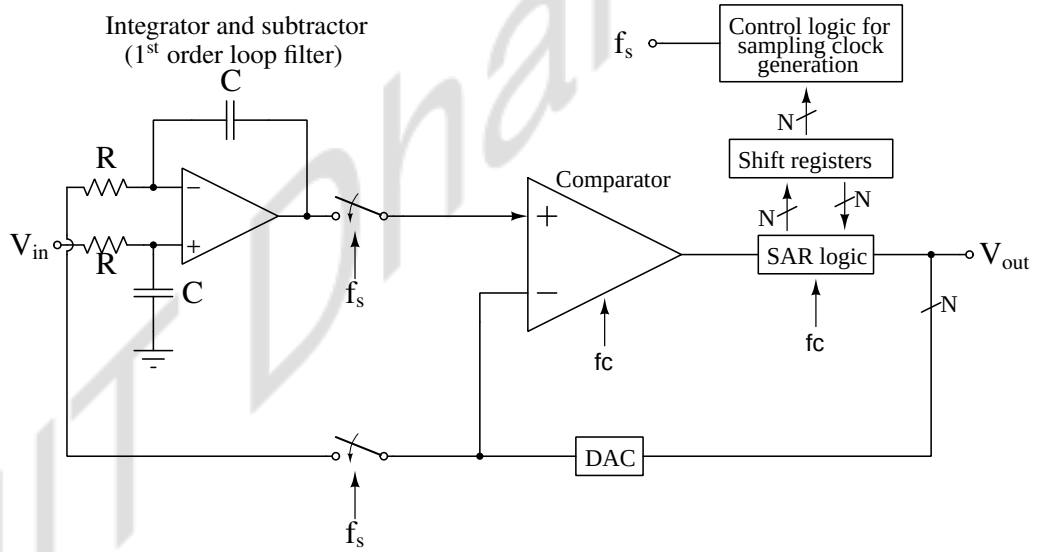


Figure 4.7: Variable-Rate 1st-order Noise-Shaping SAR ADC

4.1.2 Results

Figure 4.10 compares two spectrums: conventional noise-shaping SAR ADC and variable-rate noise-shaping ADC reported in this work. Due to noise shaping, it can be observed that the noise at the lower frequencies is attenuated. Variable-rate noise shaping has a very low noise level at lower frequencies. This is due to the high oversampling ratio of the ADC, which increases the number of samples sampled into the ADC, thereby improving the signal quality by

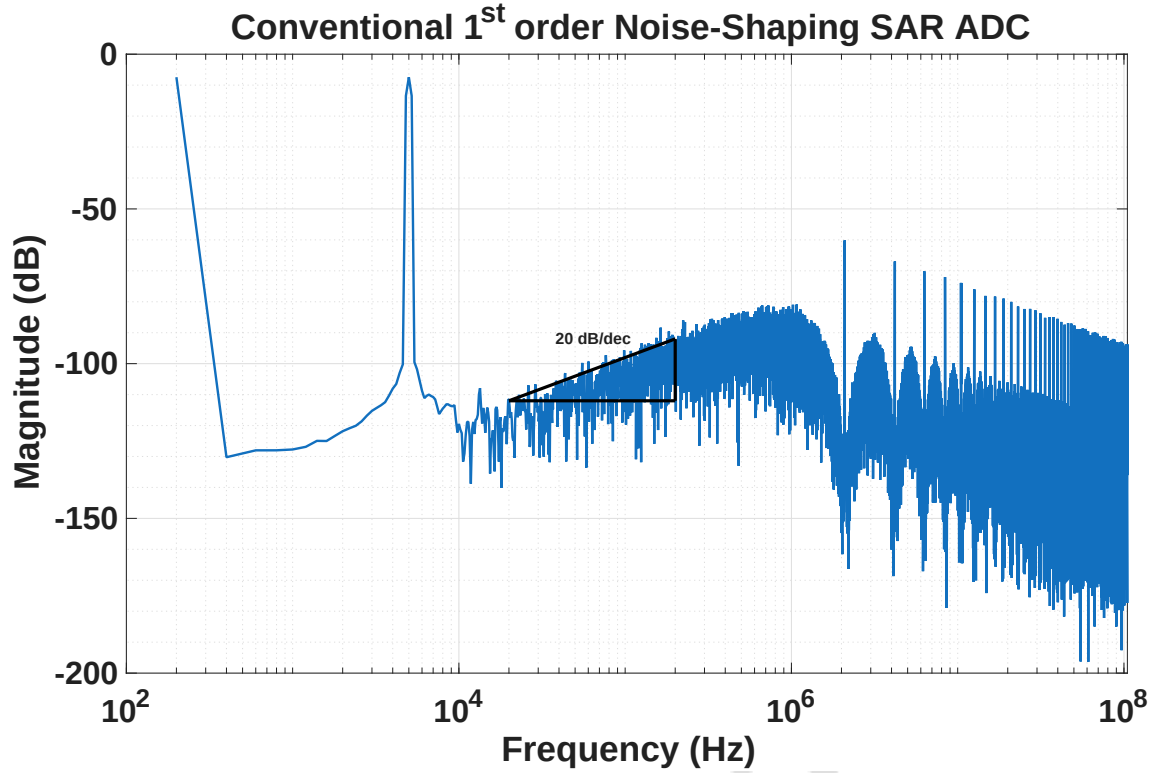


Figure 4.8: Spectrum of Conventional 1st order noise-shaping SAR ADC

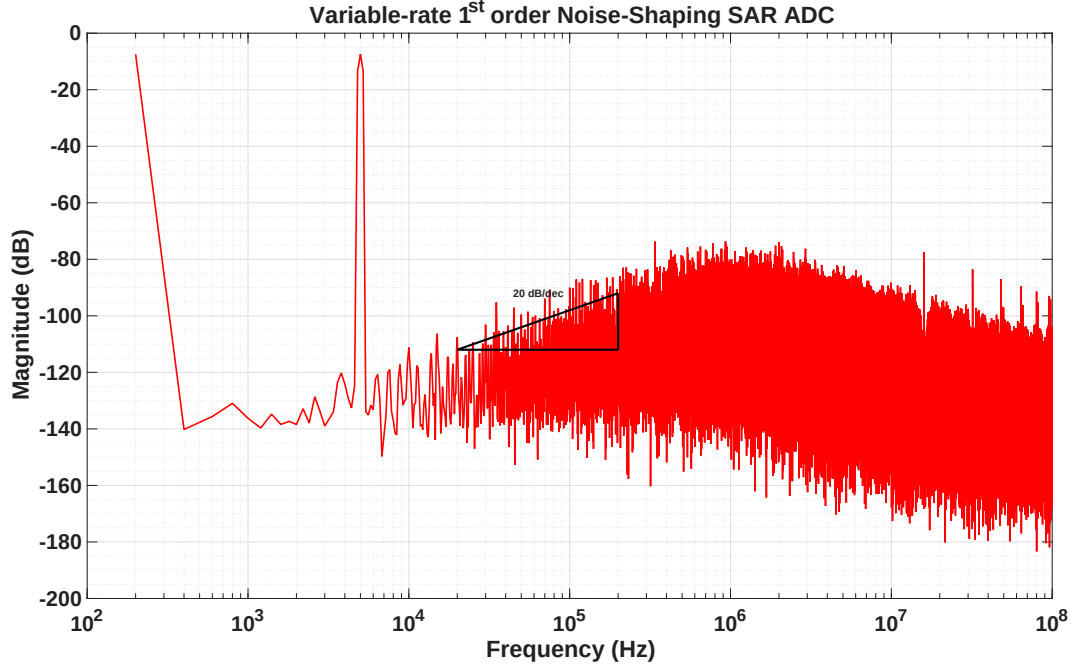


Figure 4.9: Spectrum of Variable-rate 1st order noise-shaping SAR ADC

reducing the noise levels. These noise levels are further attenuated due to the delta-sigma loop, and hence the noise level is for the variable-rate ADC. Out-of-band Noise in the first-order noise shaping architecture will have a slope of 20dB/decade. It is evident from Figures 4.8 and 4.9

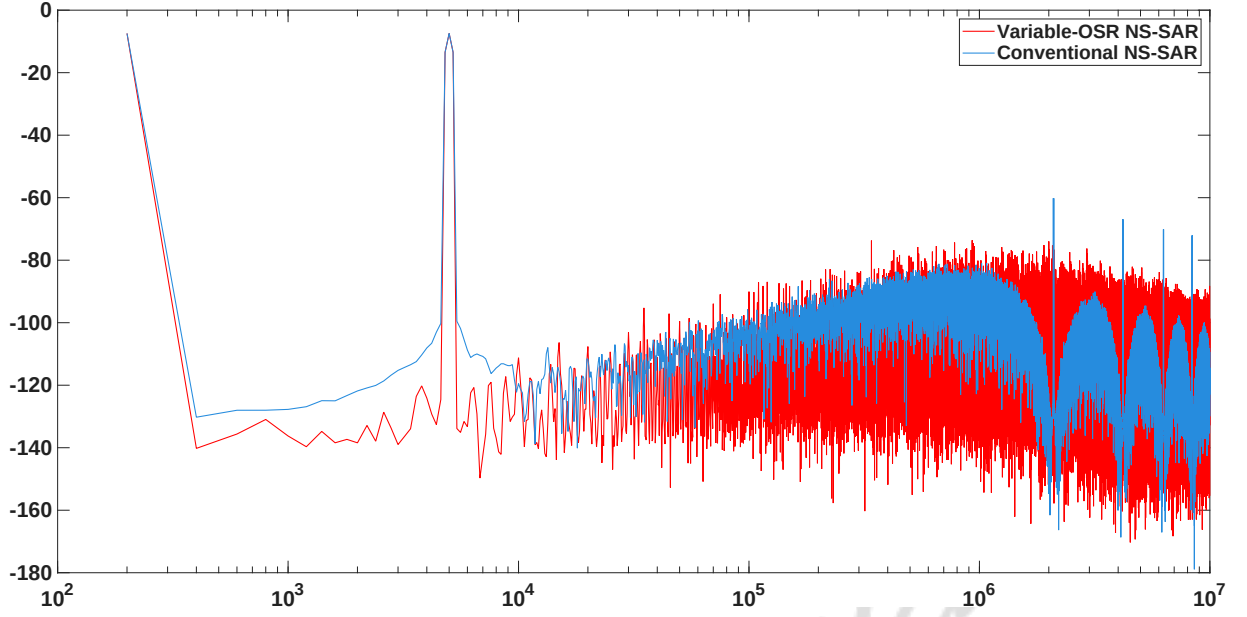


Figure 4.10: Spectrum of Conventional vs Variable-rate 1st order noise-shaping SAR ADC

that both spectra are due to first-order noise-shaping as the slope is 20dB/decade. Noise shaping ADC consists of nulls in the spectrum whose position is dependent on the sampling frequency, as shown in Figure 4.8. Additionally, these nulls reduce the noise levels of the frequency components outside the band. In the Variable-rate ADC, the positions of nulls will not be visible as shown in Figure 4.9. This is because the sampling frequency varies dynamically, causing the positions of such spectral nulls to shift over time and thereby making them less distinct in the averaged spectrum. These noise-shaping advantages can be extended to higher orders or multi-stage architectures. This will decrease the noise levels further, thereby increasing the effective signal-to-noise ratio improvement. The slope of the SNR vs order of DSM and SNR vs number of stages plots is higher for the variable-rate noise-shaping ADC compared to the conventional noise-shaping SAR ADC. This is because of the high oversampling ratio achieved through the technique reported in this work.

4.1.3 SNR and ENOB

- Frequency of the input signal is 5KHz
- Frequency of the controller clock signal is 16MHz.

An 8-bit Conventional SAR ADC takes 8 cycles per conversion irrespective of the input signal. The implemented 8-bit variable-rate LSB-First SAR ADC architecture takes 3.5 cycles

per conversion for the above specified test signals, leading to an improvement in the effective OSR. This effective improvement in OSR will improve the SNR by reducing the in-band noise, which can be observed in Figure 3.17. Improvement in the SNR of the ADC compared to conventional SAR ADC for the above test signals is 10dB in case 1st order noise-shaping. The SNR improvement is equivalent to ENOB improvement of 1.6 bits.

Another way to improve the noise-shaping at lower frequencies is by improving the order of DSM. This will introduce a variation in the roll-off of the loop filter, due to which the noise-shaping of lower frequency components increases. Design of 2nd order DSM is discussed in the upcoming section.

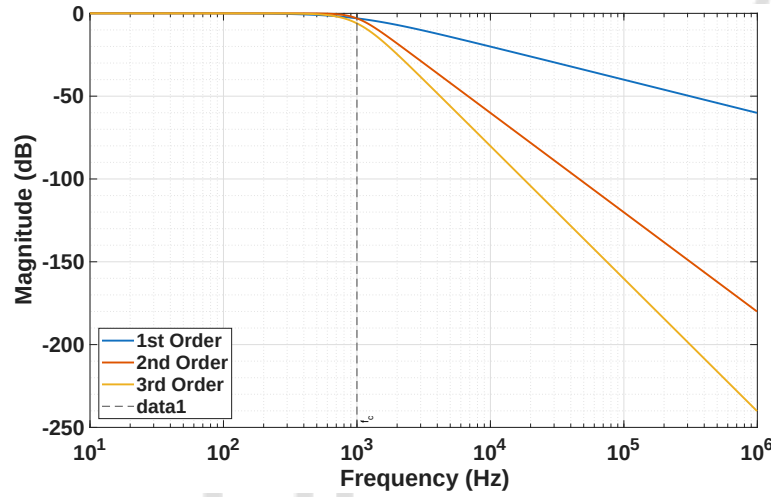


Figure 4.11: Signal Transfer Function response for DSM of different orders

4.2 2nd order $\Delta\Sigma$ ADC

The block diagram 2nd order noise shaping ADC is shown in Figure 4.12. The output is given by

$$V_{\text{out}} = \frac{L_1(s)L_2(s)}{1 + L_2(s) + L_1(s)L_2(s)} V_{\text{in}} + \frac{1}{1 + L_2(s) + L_1(s)L_2(s)} Q_e$$

where,

Q_e is quantization noise

Cascade Integrator Feedback (CIFB) architecture consists of local feedback for each integrator, shown in Figure 4.12. This architecture is preferred as it will have local feedback, which will help improve the speed of the architecture.

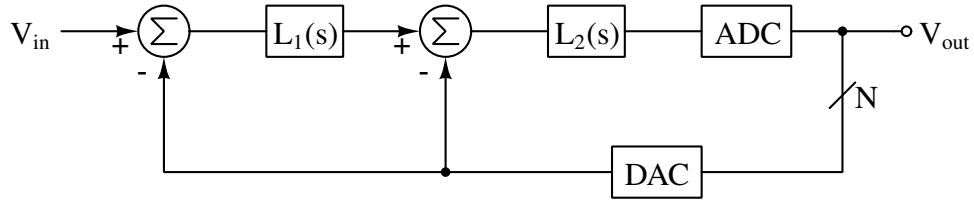


Figure 4.12: 2nd order noise shaping ADC

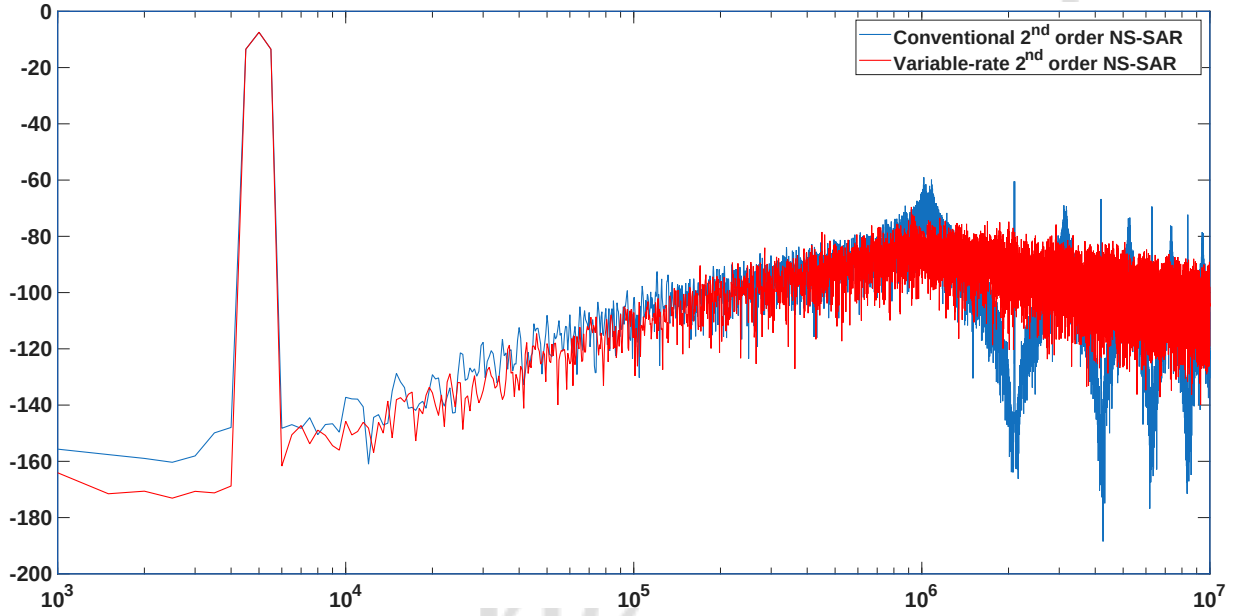


Figure 4.13: Spectrum of 2nd order Conventional SAR ADC v Variable-rate SAR ADC

While SNR of MOD1 improves by 9dB when the sampling frequency is doubled, MOD2 increases by 15dB. Results of conventional noise shaping SAR ADC and this work, i.e., variable-rate noise shaping SAR ADC, are compared; the improvement can be seen in Figure 4.13.

- Frequency of the input signal is 5KHz
- Frequency of the controller clock signal is 16MHz.

An 8-bit Conventional SAR ADC takes 8 cycles per conversion irrespective of the input signal. The implemented 8-bit variable-rate LSB-First SAR ADC architecture takes 3.5 cycles per conversion for the above specified test signals, leading to an improvement in the effective OSR. This effective improvement in OSR will improve the SNR by reducing the in-band noise, which can be observed in Figure 3.17. Improvement in the SNR of the ADC compared to conventional SAR ADC for the above test signals is 16dB in case 2nd order noise-shaping. The SNR improvement is equivalent to ENOB improvement of 2.6 bits.

4.3 Consequences of Non-Uniform sampling over uniform sampling

Uniform sampling, governed by the Nyquist–Shannon sampling theorem, provides a well-defined framework for perfect signal reconstruction when the sampling rate exceeds twice the highest frequency component. In contrast, non-uniform sampling—where sampling instants are irregular—breaks this structured periodicity in time, leading to fundamentally different spectral characteristics. This section examines the consequences of non-uniform sampling relative to uniform sampling observed in this work.

Advantages:

1. Increased speed and ADC will be adaptive to the input signal as sampling is event-driven.
2. Increased speed gives room for sampling more points in a given time frame, thereby improving SNR.
3. Real-world signals are not single tone, and the signal contains noisy components as well, which alias back. Non-uniform sampling reduces the aliasing effect if not eliminate completely.

Disadvantages:

1. The reconstruction of the signal becomes complex.
2. The absence of nulls highlights the increase in the outerband noise.

Chapter 5

Conclusion and Future Work

5.1 Conclusion

This thesis has presented the modeling, analysis, and evaluation of a noise-shaping successive approximation register (NS-SAR) analog-to-digital converter, targeting high-resolution data conversion with improved bandwidth. The thesis also reviews the LSB-First SAR ADC, a variant of the SAR ADC used in applications where the input signal shows moderate or low activity. This architecture has been modified to enhance the bandwidth and SNR.

This thesis proposes a non-uniform sampling technique to enhance the SNR of the SAR ADC. By introducing the LSB-First SAR ADC as a quantiser in the $\Delta\Sigma$ loop, the SNR is increased further. This was validated through behavioral simulations demonstrating a substantial improvement in in-band SQNR compared to a conventional SAR ADC operating at the same ADC clock.

The architecture is tested for different input signal bandwidth and DSM orders. The results were compared with conventional SAR ADC, and the improvement in SNR is observed. The 8-bit Variable-rate SAR ADC was tested with an input signal frequency of 5KHz and ADC clock (SAR controller clock) of 16MHz. Cycles per conversion decreased from 8 cycles to 3.5 on average in variable-rate SAR ADC compared with conventional SAR ADC. This led to improvement in effective OSR, thereby improving SNR by 3dB. SNR improvement increased to 10dB and 16dB with the use of 1st and 2nd order loop filters, respectively, using variable-rate SAR ADC as quantiser.

The findings of this thesis demonstrate a Variable-rate NS-SAR ADCs which improves the Signal-to-Quantization Noise Ratio (SQNR) without trading the input signal bandwidth. The

results of the same has been tabulated below :

Input signal frequency= $f_{in}=5$ KHz

SAR clock frequency= $f_c=16$ MHz

Architecture	Noise Shaping	Order	SNDR (dB)
SAR ADC	No	-	73dB
Variable OSR LSB-First SAR ADC	No	-	76.5dB
SAR ADC	Yes	1	108.6dB
Variable OSR LSB-First SAR ADC	Yes	1	116.4dB
SAR ADC	Yes	2	140.5dB
Variable OSR LSB-First SAR ADC	Yes	2	156.6dB

5.2 Future work

5.2.1 Design of Decimator for variable-rate ADC

It is necessary to have a decimation filter for the reconstruction and downsampling of an over-sampled signal. The complete ADC block diagram, including filtering and downsampling, is shown in Figure 5.1.

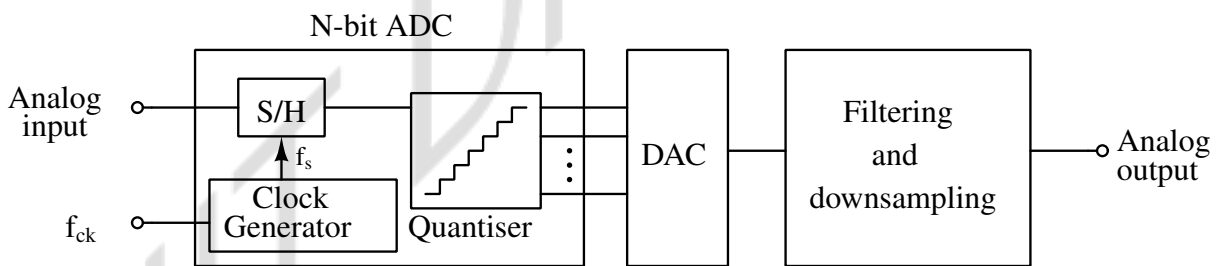


Figure 5.1: Block Diagram of ADC architecture including Quantization, reconstruction, and downsampling

Reconstruction of the signal is done using the filter. These days, digital filters are preferred over analog filters for reconstruction due to the following.

1. Better noise immunity and signal integrity
2. Better stability, precision, and performance
3. Easy to design, implement, and digital can implement advanced features

A digital filter depends on the current sample and a finite number of past samples for signal reconstruction. The output of a digital filter is given by

$$y[n] = \sum_{k=0}^N c[k] x[n-k] \quad (5.1)$$

where:

1. $c[k]$ = filter coefficients
2. $x[n-k]$ = Input samples
3. $N+1$ = Number of taps in the filter

The filter stores the previous inputs using the delay elements and adds them all after multiplying them with the filter coefficients, as shown in Figure 5.2.

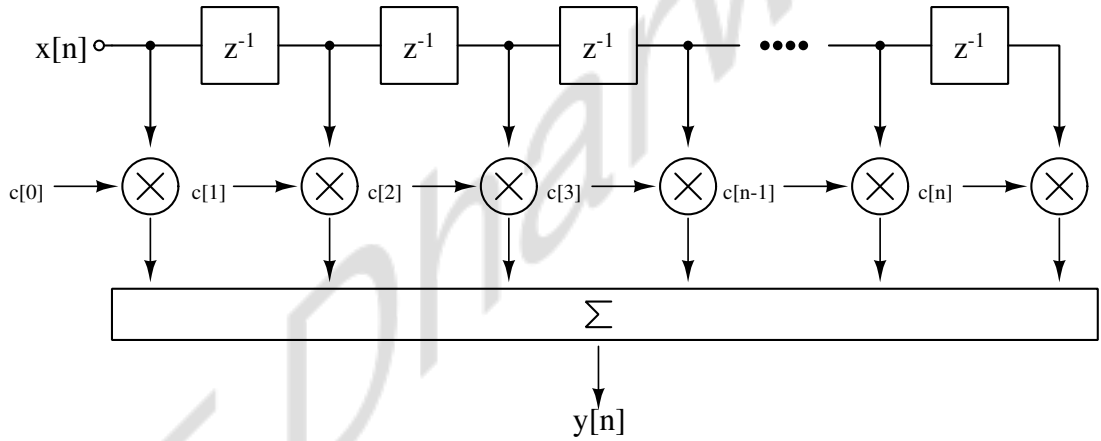


Figure 5.2: Block Diagram of Finite Impulse Response (FIR) filter for Linear-Time Invariant (LTI) system [5]

Conventional ADC architecture does fixed-rate sampling. So, a digital filter can be implemented with delay elements and coefficients as shown in Figure 5.2 to adjust the response of the filter so that the harmonics can be attenuated properly.

This work described a non-uniform sampling ADC, and sampling instances are not predictable. Consecutive samples will be at non-uniform intervals, and hence a conventional digital filter cannot be used. FIR filters are essentially discrete-time LTI systems, and this architecture is not one of them.

To reconstruct the signal, an analog filter should be used, or interpolate non-uniform samples, create uniform samples, and then use a conventional digital filter. Otherwise, it is necessary

to build a filter that will adapt according to the sampling clock. This will go down as one of the future works.

5.2.2 Digital-to-Analog Converter: Design choices

Conventionally, SAR ADC uses Charge-Sharing DAC in the feedback in the architecture due to its zero static power dissipation nature. If the LSB-First SAR ADC is considered, it can be observed that its MSB capacitors stay in the same states for longer periods since the signal is oversampled. Due to the presence of parasitics, there will be a possibility of charge leakage, which will affect the ADC performance.

A possible solution would be to reset the capacitors periodically. Another possible solution would be to explore the performance of the ADC with other DAC architectures.

Appendix A

Spectrum Analysis

This chapter discusses the spectral analysis of uniform and non-uniform sampled ADC. The objective is to analyze how the sampling strategy influences the frequency-domain representation of the signal.

A.1 Uniform Fast Fourier Transform (FFT)

In this section, the spectrum of a signal acquired through conventional uniform sampling is examined. Every sample is equally spaced due to a fixed sampling frequency. In such cases, FFT is used to compute the Discrete Fourier Transform (DFT) of a sequence. It transforms a signal from the time domain to the frequency domain, revealing the spectral content such as frequency components, harmonics, and noise.

Consider a discrete-time signal $x[n]$ of length N ; the following mathematical equation defines its DFT [8]:

$$X[k] = \sum_{n=0}^{N-1} x[n] \cdot e^{-i \frac{2\pi}{N} kn}, \quad k = 0, 1, \dots, N-1 \quad (\text{A.1})$$

Here:

1. $N = \frac{f_s}{f_{in}}$ is number of samples.
2. $x[n]$ is the input signal sampled at uniform frequency f_s
3. f_{in} is the input signal frequency
4. $X[k]$ is the frequency-domain representation

5. k is the number of frequency bins

6. i is the imaginary unit

So far, spectrum analysis for the cases where the sampling rate is fixed has been discussed. In the upcoming section, we will be discussing the FFT of the signal when the sampling rate is not uniform/fixed.

A.2 Non-Uniform Discrete Fourier Transform (NU-DFT)

Non-Uniform sampling consists of data sampled at non-uniform intervals. In such cases, the sampling frequency can vary over time, making it impossible to use the FFT for spectrum analysis. NU-DFT will be used in such cases, as detailed in this section. This section details the spectrum analysis of non-uniformly sampled signals [8].

Given a set of non-uniformly spaced time samples $\{t_n\}$ and corresponding signal values $\{x_n\}$, the NUDFT is defined as:

$$X_k = \sum_{n=0}^{N-1} x_n e^{-2\pi i k t_n f_{in}}, \quad k = 0, 1, \dots, K-1 \quad (\text{A.2})$$

This contrasts with the standard DFT, which assumes a uniform time spacing as shown in Equation A.1.

Here:

- x_n is the input signal sampled at non-uniform times t_n
- f_{in} is the input signal frequency
- X_k is the frequency-domain representation
- K is the number of frequency bins
- i is the imaginary unit

A.3 Testing setup

- The signal frequency and the controller clock frequency are made to be mutually prime to collect a unique set of data points on each cycle and make the quantization noise non-periodic to reduce harmonics [9].

- Number of FFT points should be at least $\frac{f_c}{2f_{in}}$ (f_c is the frequency of the controller clock) as highest possible sampler clock frequency is $\frac{f_c}{2}$ [9].
- Windowing is necessary to reduce spectral leakage. This is because the signal does not appear periodic in the observation window and leads to discontinuity and leakage, which will add extra frequency components [9].

IIT Dharwad

Appendix B

Design of 12-bit SAR ADC for Tire Pressure Monitoring Systems (TPMS)

B.1 Introduction

TPMS has become an integral part of modern automotive safety with the growing demand and requirement for intelligent vehicular diagnostics. A TPMS continuously monitors the air pressure inside tires and delivers timely alerts to the vehicle control system to prevent hazardous situations. It consists of sensors to detect the pressure and temperature variations, whose output should be processed further into the digital domain for further analysis. Since the sensor outputs are analog in nature, it is necessary to have a block that can convert an analog signal to digital. This conversion is supposed to be low-noise, power-efficient, and area-efficient.

Among the numerous ADC architectures available which were discussed in Chapter 1, the Successive Approximation Register (SAR) ADC has in particular has emerged as a preferable solution for compact, battery-powered sensing applications such as TPMS. This choice is mainly due to the SAR ADC's excellent energy efficiency, moderate speed, scalability and a digital controller, which enables long-term operation without frequent battery replacements. As TPMS is expected to operate for a longer duration on a small coin cell, power consumption becomes the most important design constraint, due to which SAR ADCs are preferred over power-hungry architectures like flash or pipelined ADCs. Also, considering the bandwidth requirements for TPMS Flash and pipeline architectures becomes an overkill.

A typical TPMS architecture consists of a pressure sensor followed by analog signal conditioning, an ADC, a digital processing unit, and a wireless transmitter. The pressure signal and

temperature data generated by the sensors are typically slowly varying, low-bandwidth analog data that must be digitized accurately. These favor the use of SAR ADCs, which offer high resolution (10–14 bits), sufficient sampling rates (typically 10–200 kS/s for TPMS), and excellent power performance [1].

The ADC should not be sensitive to temperature variations in automotive environments which is mostly ranging from -40°C to $+140^{\circ}\text{C}$. Also, the sensitivity to power supply variations should be minimal for efficient performance. Therefore, the SAR ADC must be designed to exhibit robust performance across PVT (process, voltage, temperature) variations.

The SAR ADC architecture consists of a sample-and-hold circuit, a capacitive DAC (CDAC), a dynamic comparator, and a successive approximation control logic. The conversion starts with sampling the signal onto the CDAC array, followed by a binary-search comparisons that rapidly converge the digital output to required state by an iterative method as discussed in Section 2.2.2.

Static state performance of the designed ADC is crucial for TPMS application, as it is supposed to be in sleep or inactive state at times, as the TPMS modules are involved in low-duty-cycled operation. The goal of the design is to achieve the required resolution and accuracy listed in Table B.1.

Parameters	Requirement
V_{ref}	1.5V
Resolution	12 bits
ENOB	11 bits
Max INL	1LSB
Max DNL	1 LSB
Throughput	1MSPS
SINAD	70dB @ 10KHz

Table B.1: ADC specifications for TPMS

This work focuses on the design and implementation of SAR ADC for Tire Pressure Monitoring Systems.

B.2 Implementation and Design

This work implements a fully differential SAR ADC as shown in Figure B.1. A fully differential SAR ADC is preferred over a single-ended architecture for several reasons:

1. **Suppression of common-mode noise and even-order distortion:** Noisy components that appear on both lines as common mode noise are inherently canceled by differential structure and will also be impactful in mitigating the even-order harmonic distortion. This leads to improved linearity and better spectral performance, which is critical for high-resolution data conversion.
2. **Improved dynamic range:** Linearity is a major concern in high-resolution data conversion. A fully differential architecture support a larger effective signal swing while maintaining the signal integrity. This increases the available headroom, effectively increasing the dynamic range.

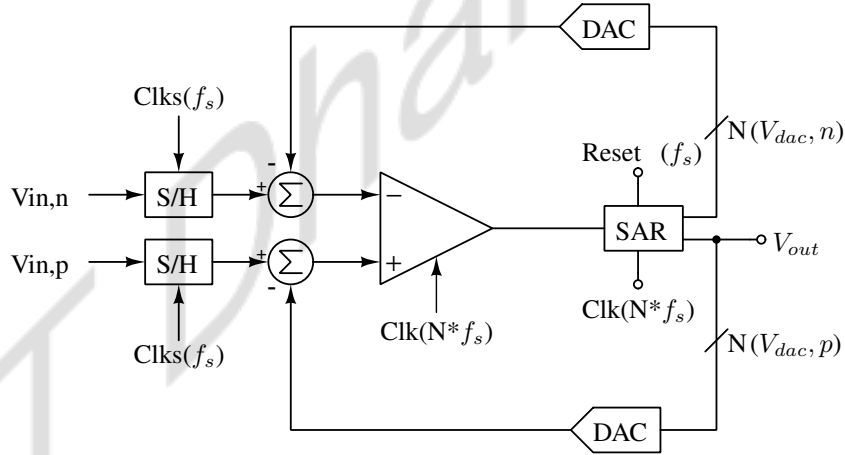


Figure B.1: Block diagram of fully differential 12-bit SAR ADC

The following section focuses on the design and implementation of a 12-bit SAR ADC with a reference voltage of 1.5V.

B.2.1 Design of dynamic comparator

Sense amplifier based comparator is used in the design as shown in Figure B.2(a). It has 2 phases of operation: the precharge phase and the decision phase. In the precharge phase, clock $\text{Clk}=0$, which charges the parasitic capacitors at $\bar{r}$ and $\bar{s}$ nodes to V_{dd} through reset transistors

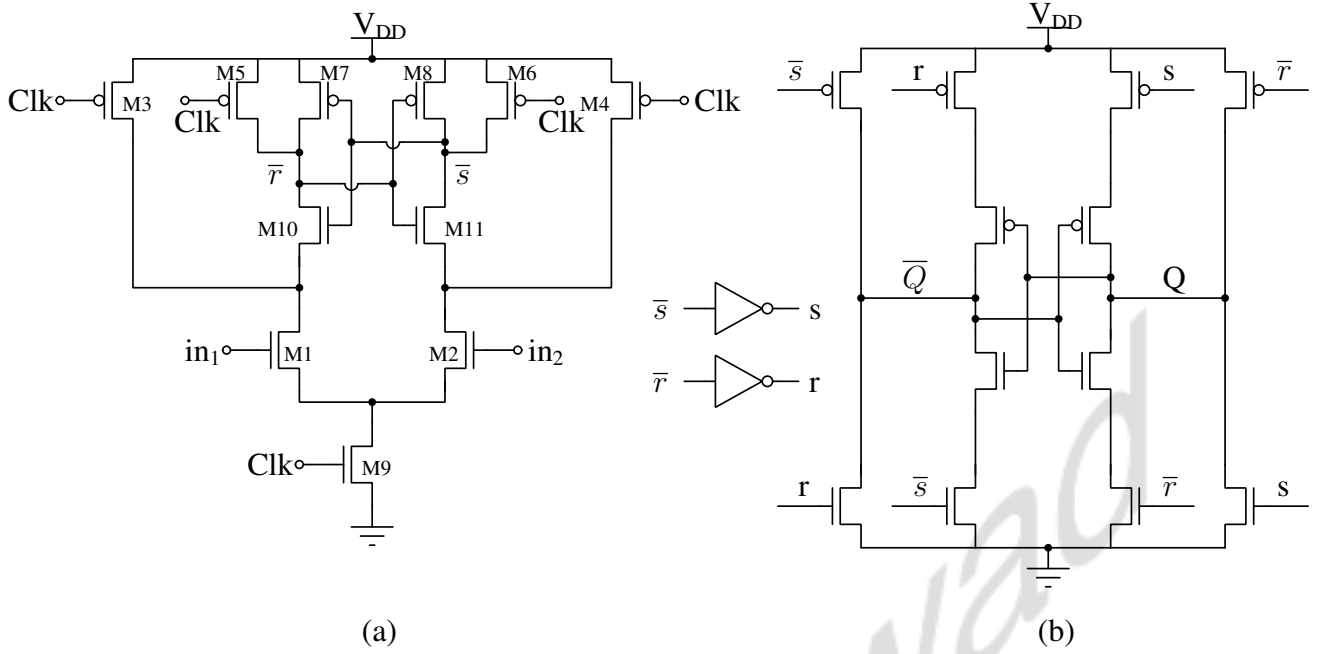


Figure B.2: Strong-arm Latch based Comparator [10]

M5 and M6 as shown in Figure B.3. This phase ensures that both the nodes are charged to identical voltages, which can eliminate memory effects from previous conversion cycles and also maximize the sensitivity to input.

When $Clk=1$, the comparator enters the decision phase. In this phase, transistor M9 will be turned on and tail current will flow through transistors M1 and M2. The input voltage applied to in_1 and in_2 determines the relative conduction of M1 and M2. If $in_1 > in_2$, the current in transistor M1 will be greater than that in M2. Due to this, the parasitic capacitor at the node $\bar{r}$ discharges faster than $\bar{s}$ as shown in Figure B.3. At one point, $\bar{r}$ node voltage drops below the threshold voltage of nmos, due to which M11 will be turned off and M8 will be turned on (M10 will be turned off if $in_1 < in_2$). Current will no more flow through M2. $\bar{r}$ will discharge to 0 while $\bar{s}$ charges to V_{dd} as shown in Figure B.3. A similar operation repeats in next clock cycle.

Since the comparator is dynamic, the output nodes $\bar{r}$ and $\bar{s}$ are reset to V_{dd} after every conversion cycle. The output is not retained. So, it is necessary to have an additional latch shown in Figure B.2(b), which can hold the output till the next clock cycle after the end of every conversion.

For the implemented design, the step size is calculated as,

$$\Delta = \frac{1.5}{2^{12}} = 366.21 \mu V$$

Any offset introduced by the comparator must be below this value to avoid incorrect bit deci-

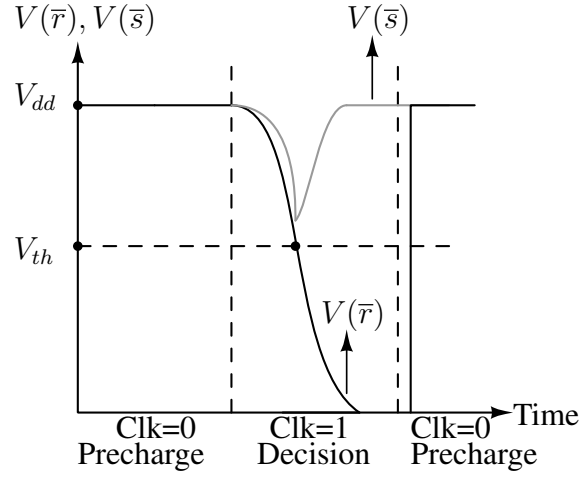


Figure B.3: Example of precharge and decision phase in sense amplifier comparator when $in_1 > in_2$

sions, missing codes, or degradation in overall linearity. In sense amplifier-based comparators, transistor mismatch—particularly in the input differential pair—can create input-referred offset.

To evaluate the offset, Monte Carlo simulations were done, which provides estimate for worst case offset in the comparator. The maximum input referred offset is observed to be 14mV as shown in Figure B.4. This is larger than the step size of the ADC, which will affect the linearity. Hence, comparator calibration is necessary.

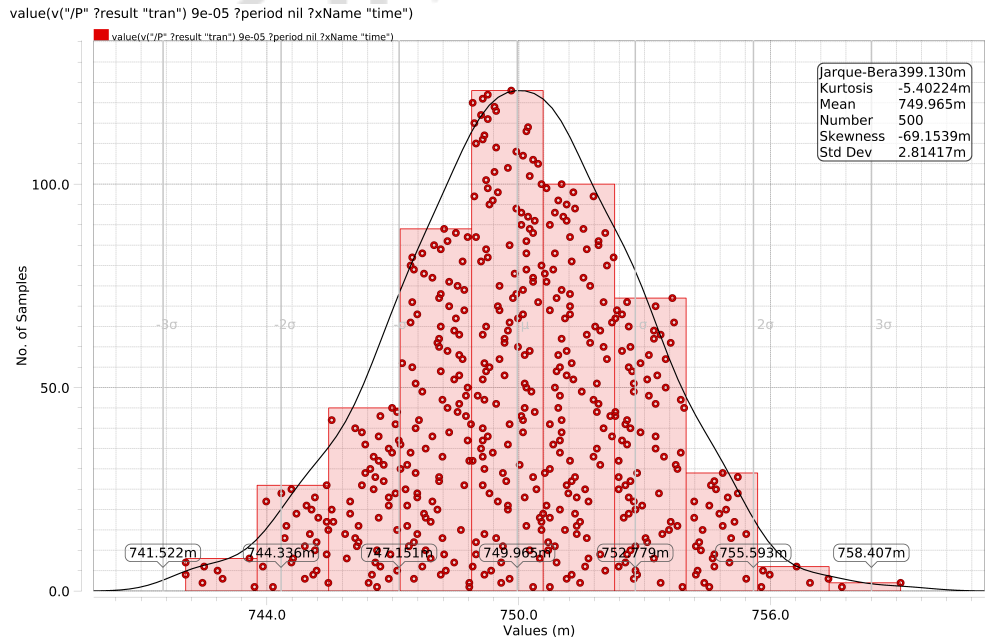


Figure B.4: Monte Carlo Simulation results before the offset calibration of Strong-Arm Latch Based Comparator

Comparator Calibration

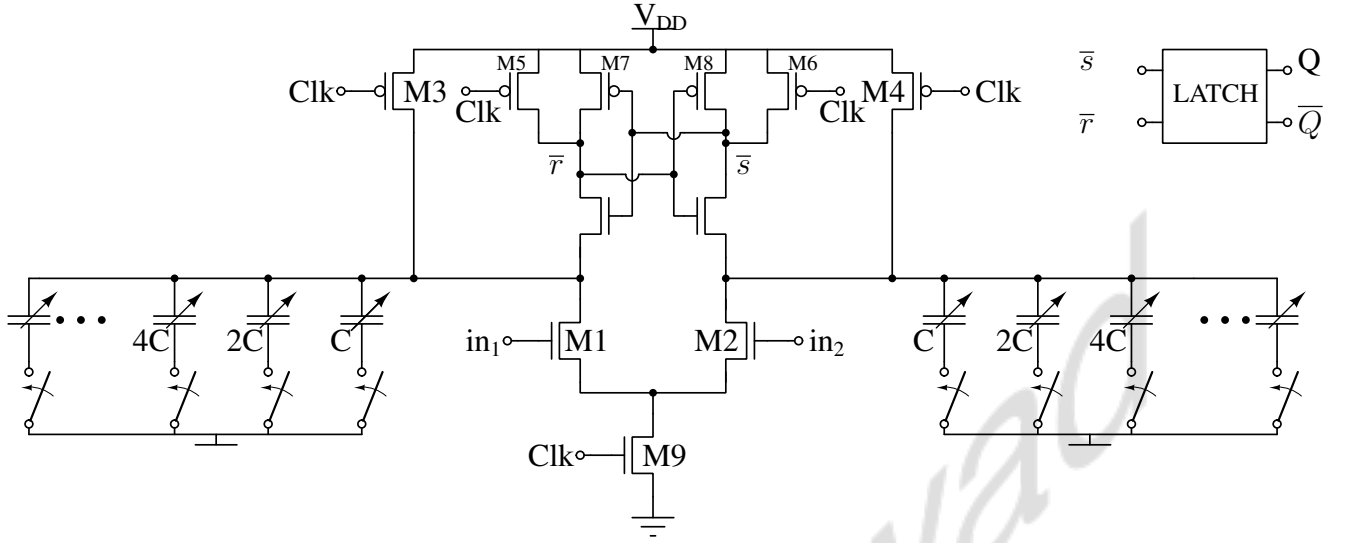


Figure B.5: Calibration of Strong-arm Latch Based comparator using capacitor array based technique [11]

During the calibration phase, in_1 and in_2 are shorted and connected to a known reference voltage. Under this condition, in ideal cases, there should be equal currents across both the input transistors. Due to the offset, the current on both sides will not be equal. So, the capacitor array is introduced additionally to balance the current by balancing the loads on both sides.

The technique used for the calibration is an iterative method. At the start of the calibration phase, all capacitors in the right array are connected to the comparator, while the left array is left disconnected. This introduces a positive offset, due to which Q will switch to V_{dd} . An up counter controls the left array, and a down counter controls the right array. As clock cycles progress, the counter incrementally increases the load on the left side while decreasing on the other side. The unit value of the load on both sides are chosen in such a way that the increment/decrement in offset on each cycle will be less than the step size of ADC. This way, the input referred offset is reduced, and once the calibration is done, Q switches from V_{dd} to 0. This is detected by a transition detector and freezes the counter outputs. Loads with different total capacitances are connected on both sides based on the amount of offset present in the comparator. These loads balance the current on both sides, thereby reducing the input referred offset. Once the calibration is completed, the shorted input nodes are connected back to the input signal.

To evaluate the worst-case offset after the calibration, a Monte Carlo simulation has been done. The worst case offset after calibration is 0.3mV as seen in Figure B.6.

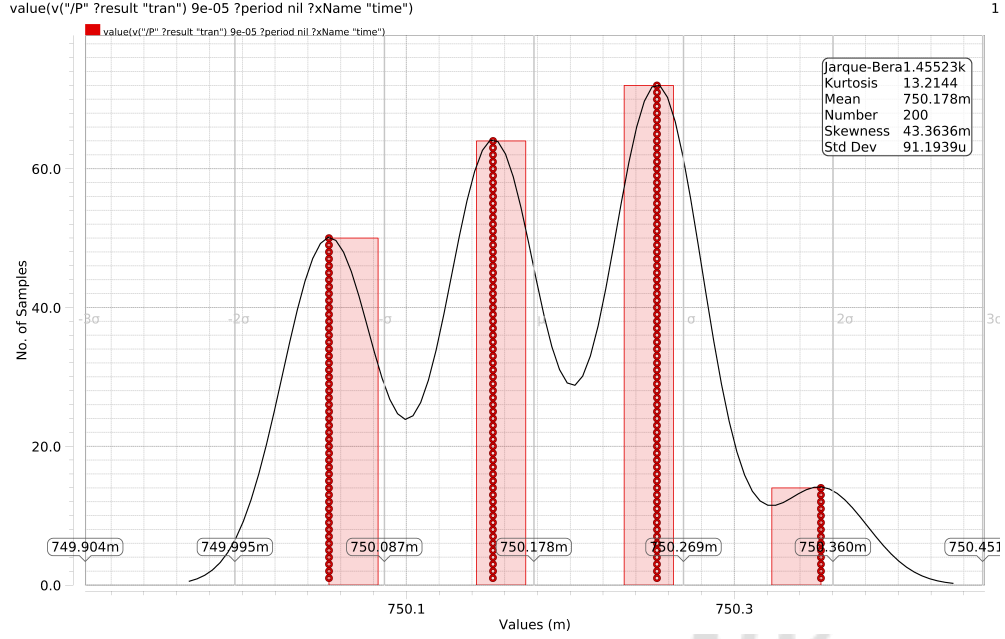


Figure B.6: Monte Carlo Simulation results after the offset calibration of Strong-Arm Latch Based Comparator

B.2.2 Design of Successive Approximation Register

Successive Approximation Register (SAR) consists of a digital logic that implements a binary search algorithm that governs the conversion process, as illustrated in Figure B.7. At the beginning of every sampling cycle, reset signal initializes the process by setting the MSB to logic 1 and rest all to 0. There after, it will iteratively converge to the required digital output, one bit at a time, based on comparator decisions while being synchronised with Clk. Timing diagram of Reset clock and SAR clock, Clk for a 12-bit SAR ADC is shown in Figure B.8.

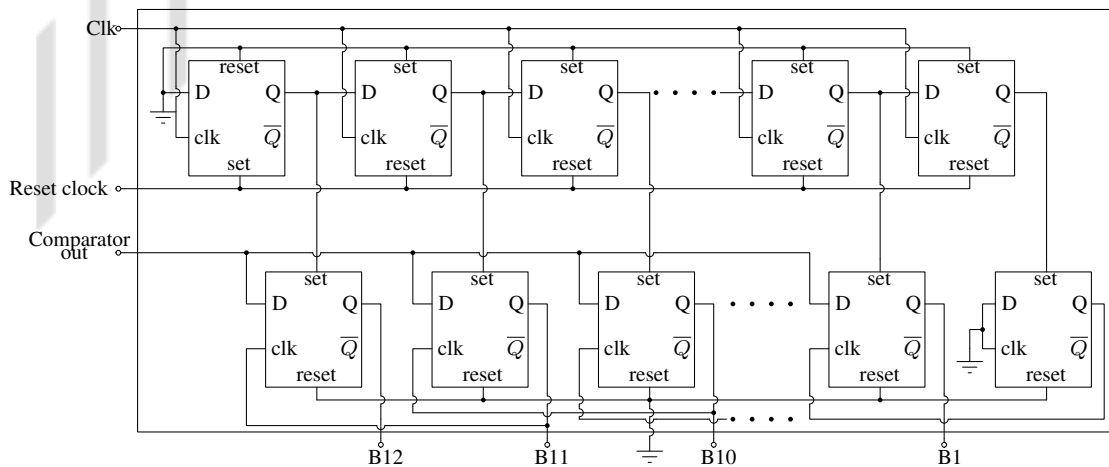


Figure B.7: 12-bit successive approximation register [2]

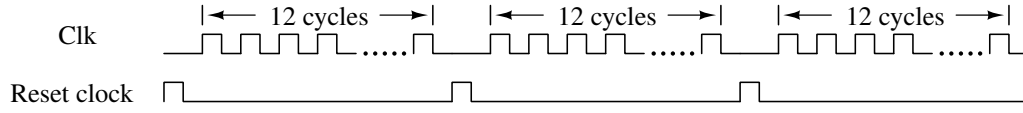


Figure B.8: Timing diagram for reset clock and SAR clock for 12-bit SAR ADC

Figure B.9 shows a schematic of a D Flip-Flop realised using a strong arm latch. It is the same design used for a dynamic comparator. This case does not require any calibration as it deals with 1's and 0's only. The latch design is modified to have set and reset pins as shown in Figure B.10.

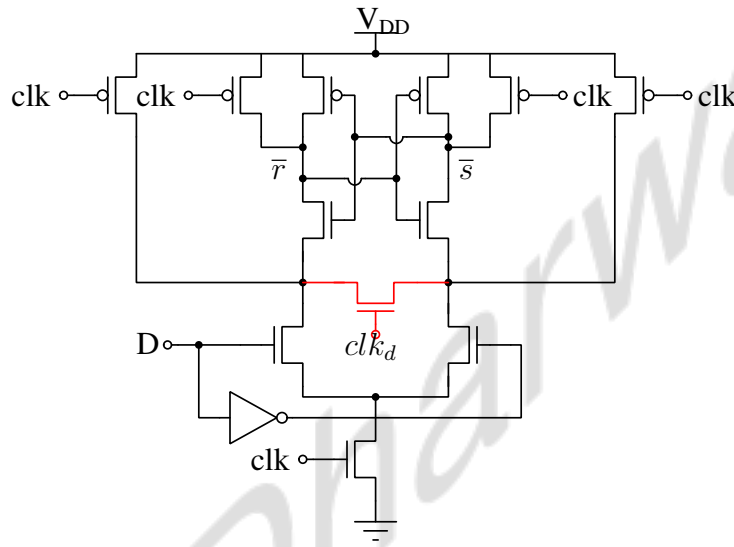


Figure B.9: Realisation of D Flip-Flop using Strong Arm Latch [10]

B.2.3 Design of charge-sharing DAC

In the implemented SAR ADC architecture, DAC is implemented using an array of binary-weighted capacitors. This kind of DAC is called Charge-sharing DAC, which is preferred in SAR ADC due to its zero static power consumption, making it energy efficient. However, conventional CDAC effects the area and speed of the ADC with the increase in resolution number of capacitors in the DAC also increases which will be an exponential growth in terms of area. This will also affect the settling time of DAC as the value of the largest capacitor is 2^{N-1} times the unit capacitor.

To mitigate these challenges, a split-array CDAC architecture is used as illustrated in Figure B.11. In this approach, DAC is partitioned into two or more smaller arrays with a bridge capacitor separating them, which is also called an attenuation capacitor. By dividing in this

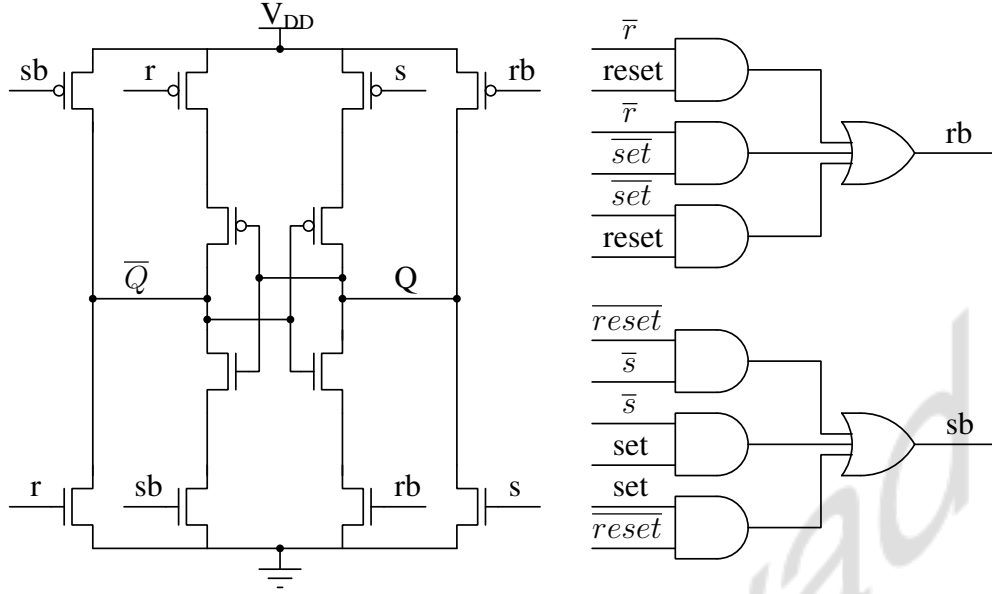


Figure B.10: D Flip-Flop latch with set and reset pins

manner, the size and value of the biggest capacitor in each array are reduced, thereby saving the area and reducing the settling time of the DAC.

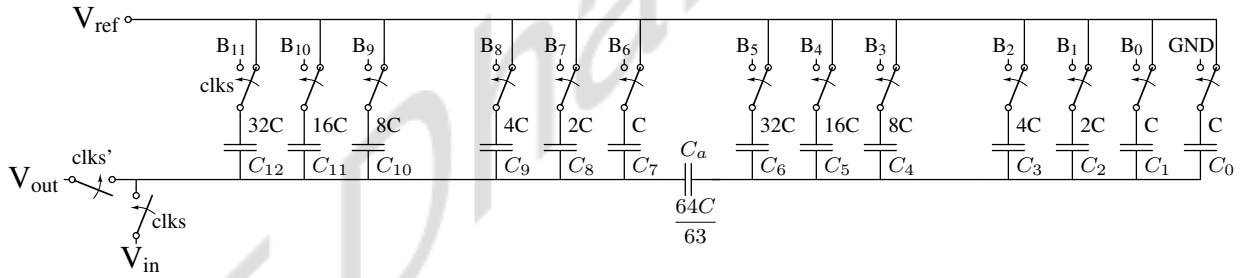


Figure B.11: 12-bit Split-Array CDAC schematic [2]

Value of bridge capacitor

Output of DAC w.r.t. Figure B.11 is given by

$$V_{out} = \frac{C_a \sum_{x=1}^6 C_x B_{x-1}}{C_a + \sum_{x=1}^6 C_x} + \sum_{x=7}^{12} C_x B_{x-1} \quad (B.1)$$

$$\frac{C_a \sum_{x=0}^6 C_x}{C_a + \sum_{x=1}^6 C_x} + \sum_{x=7}^{12} C_x$$

Equation B.1 can be used to find the value of the bridge capacitor given by

$$C_a = \frac{\text{Sum of capacitors in the left array} + \text{Unit capacitor}}{\text{Sum of capacitors in the left array}}$$

Bottom plate sampling

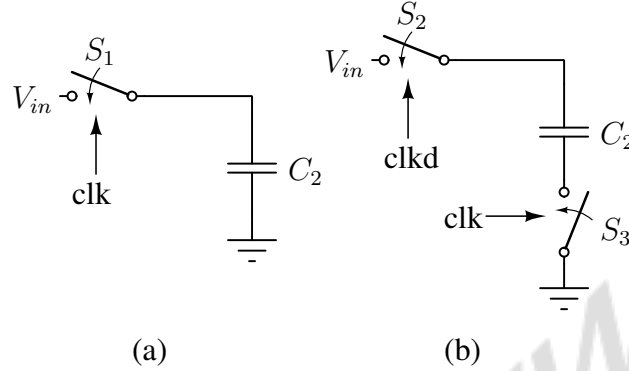


Figure B.12: (a) Conventional sampling (b) Bottom plate sampling [2]

In bottom plate sampling, an extra switch is used whose input is grounded and connected to the other end of the capacitor as shown in Figure B.12(b). Clocks fed to switches S_2 and S_3 have the same period having a small delay between each other. Switch S_3 is turned off first, and then switch S_2 . When the switch S_3 is turned off it injects a constant charge to the capacitor (Channel charge of switch S_3 will be the same all the time as input is grounded). The path towards the capacitor C_2 from switch S_2 is now a high impedance path. When switch S_2 is turned off channel charge of switch S_2 will discharge in a low impedance path, i.e., towards source V_{in} . Charge injection on C_2 is due to switch S_3 , which is fixed regardless of input V_{in} . This way charge injection effect is nullified by bottom plate sampling.

Working principle of charge-sharing DAC

CDAC is a switched capacitor circuit. Channel charge in mos leads to charge injection, which is input signal dependent. This will introduce non-linearity in the DAC, degrading the performance of ADC. Since the architecture is differential, it would be enough if the charge injection is made constant irrespective of the input, instead of nullifying it, as it would appear as a common mode signal at the comparator input. This can be done by doing bottom plate sampling as shown in Figure B.12(b).

When $\text{clk}=1$, the input signal V_{in} is sampled and stored across the capacitors. Bottom plate sampling is done to reduce the effect of charge injection due to the switches. After sampling the signal, digital bits are connected to perform $V_{in}-V_{DAC}$ operation.

B.2.4 Design of bootstrap switch

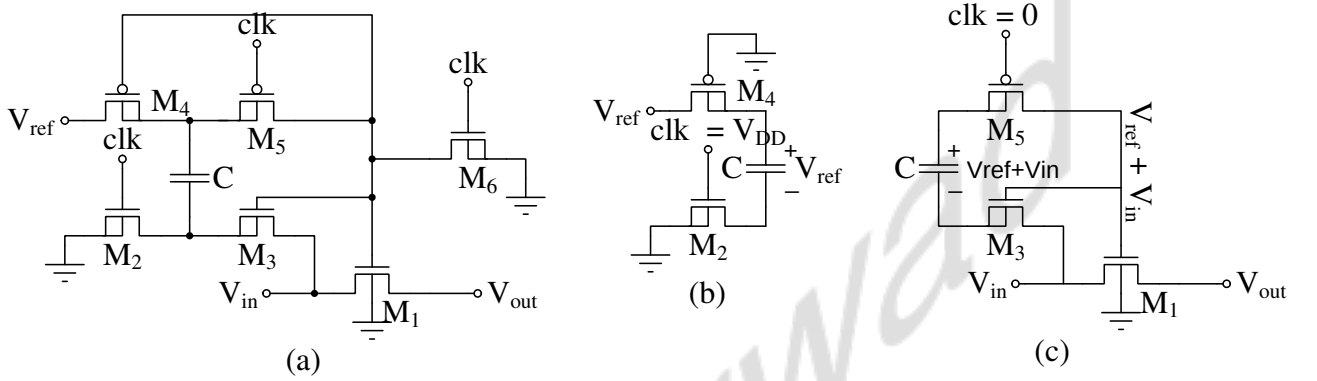


Figure B.13: (a)The Bootstrapped Switch, (b)Capacitor charging to V_{ref} when the clock is high, (c)Track Operation when the clock is low [12]

The schematic of the bootstrap switch is shown in Figure B.13(a). When $\text{clk}=1$, capacitor C charges to voltage V_{ref} as shown in Figure B.13(b). When $\text{clk}=0$, capacitor C charges to voltage $V_{ref} + V_{in}$ as shown in Figure B.13(c). In this phase, capacitor C turns on the input transistor, allowing it to pass the input. It is capable of passing strong ones and zeros.

The resistance of the MOS when the switch is ON is given by

$$R_{ON} = \frac{1}{\mu_n C_{ox} \frac{W}{L} (V_{gs} - V_t)}$$

The only varying parameter $V_{gs} - V_t$ will always be equal to V_{ref} . Hence, the switch resistance remains constant irrespective of the input v_{in} .

B.3 Results and Conclusion

The output of the differential SAR ADC is verified, and the spectrum of the ADC output was plotted, which can be seen in Figure B.14. SINAD of the ADC was calculated by giving an input signal with a frequency of 10KHz and a sampler clock with a frequency of 1MHz. SINAD and ENOB for pre-layout simulations are 70.39dB and 11.44, respectively.

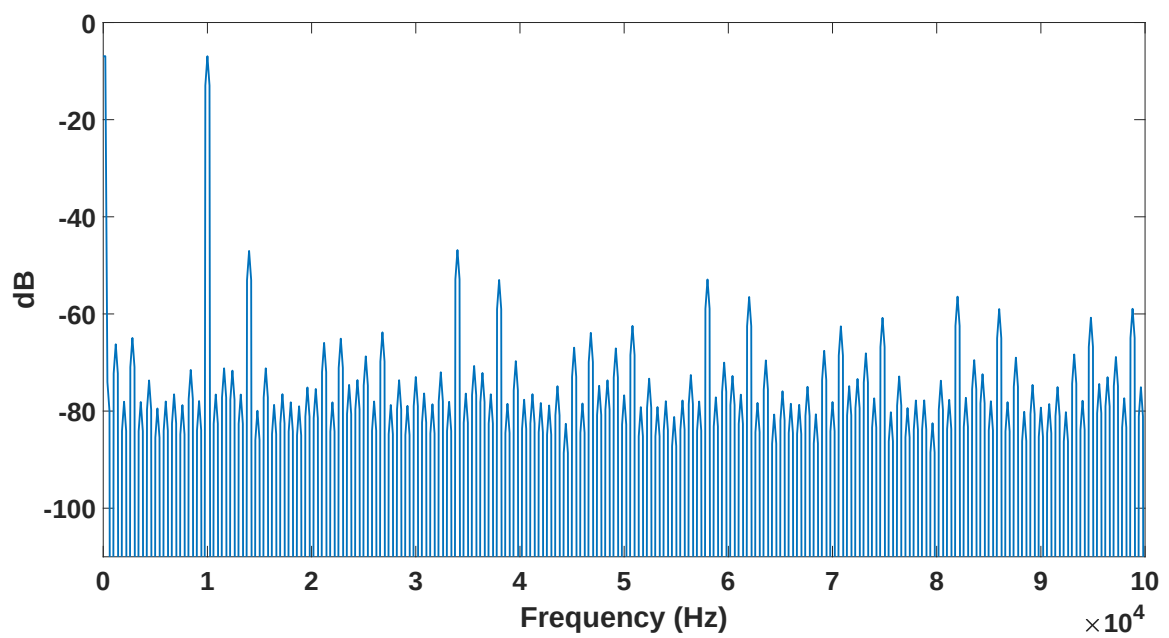


Figure B.14: Output spectrum of implemented 12-bit fully differential SAR ADC

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